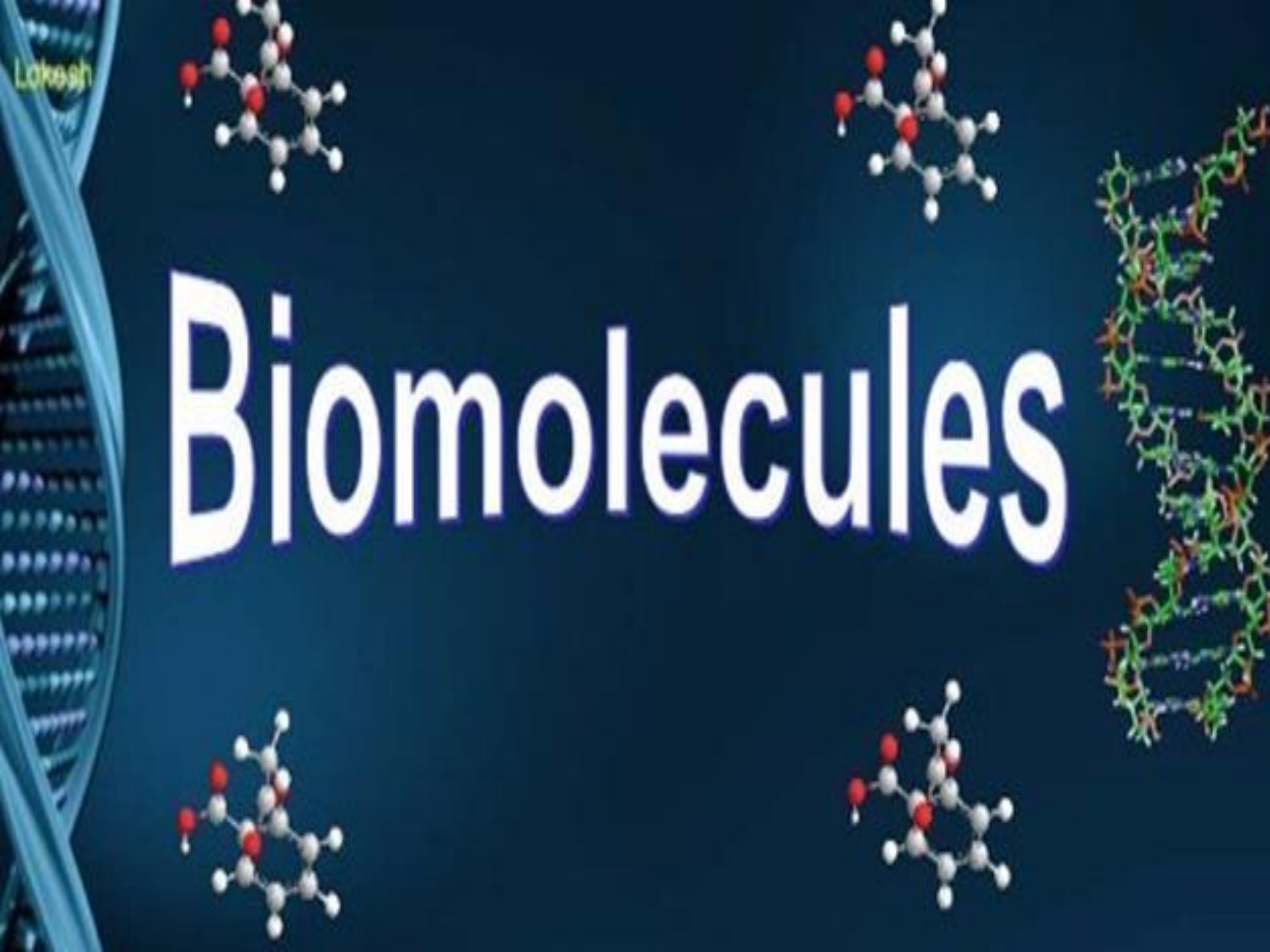

Lokesh
Biomolecules

What are BIOMOLECULES?

- **Organic compounds made by living things**
- Compounds essential to life
- Also called **biochemicals**
- Some are very large - **polymers**
- There are thousands of different biomolecules, but are **separated into 4 categories**

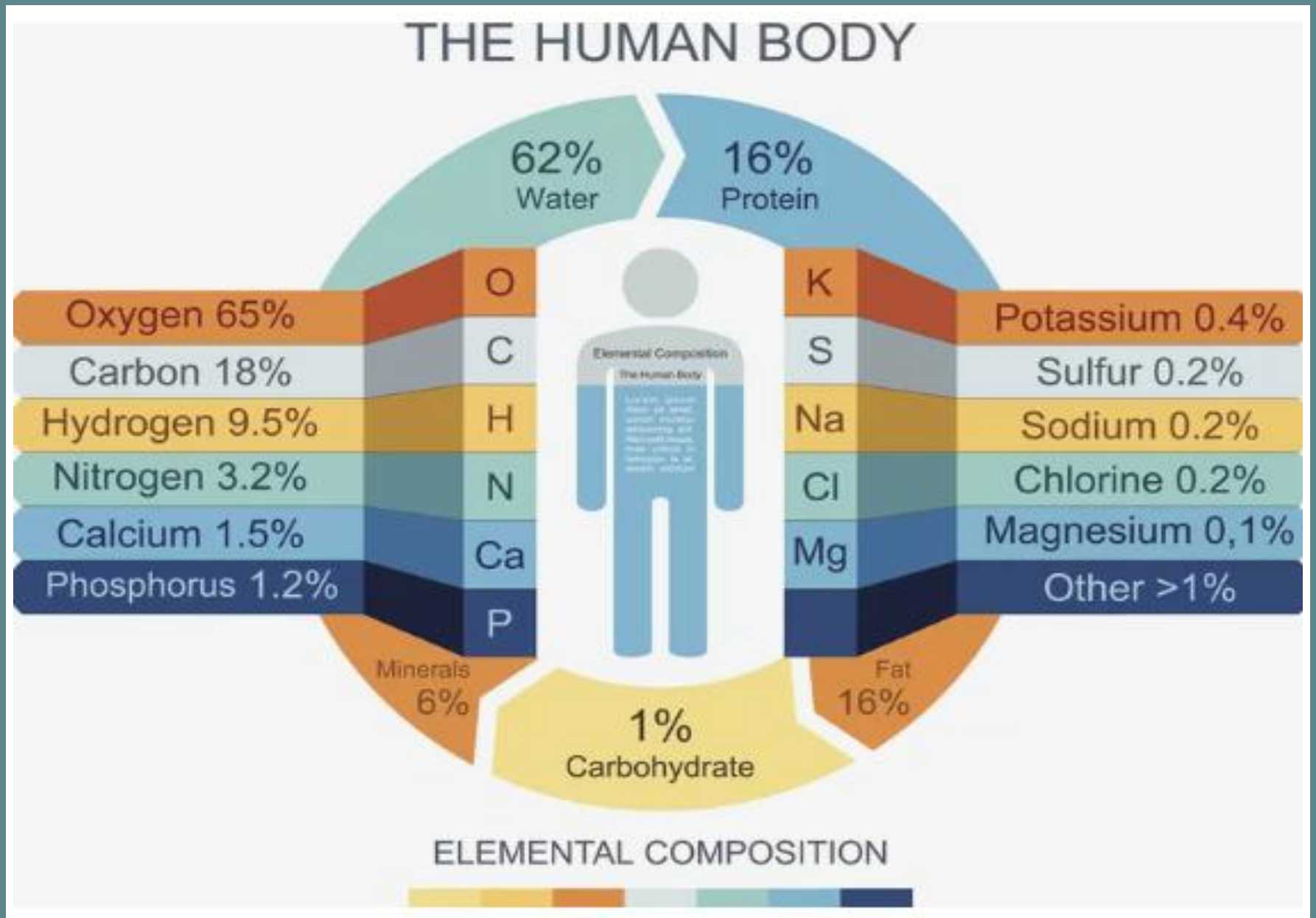


Carbohydrates

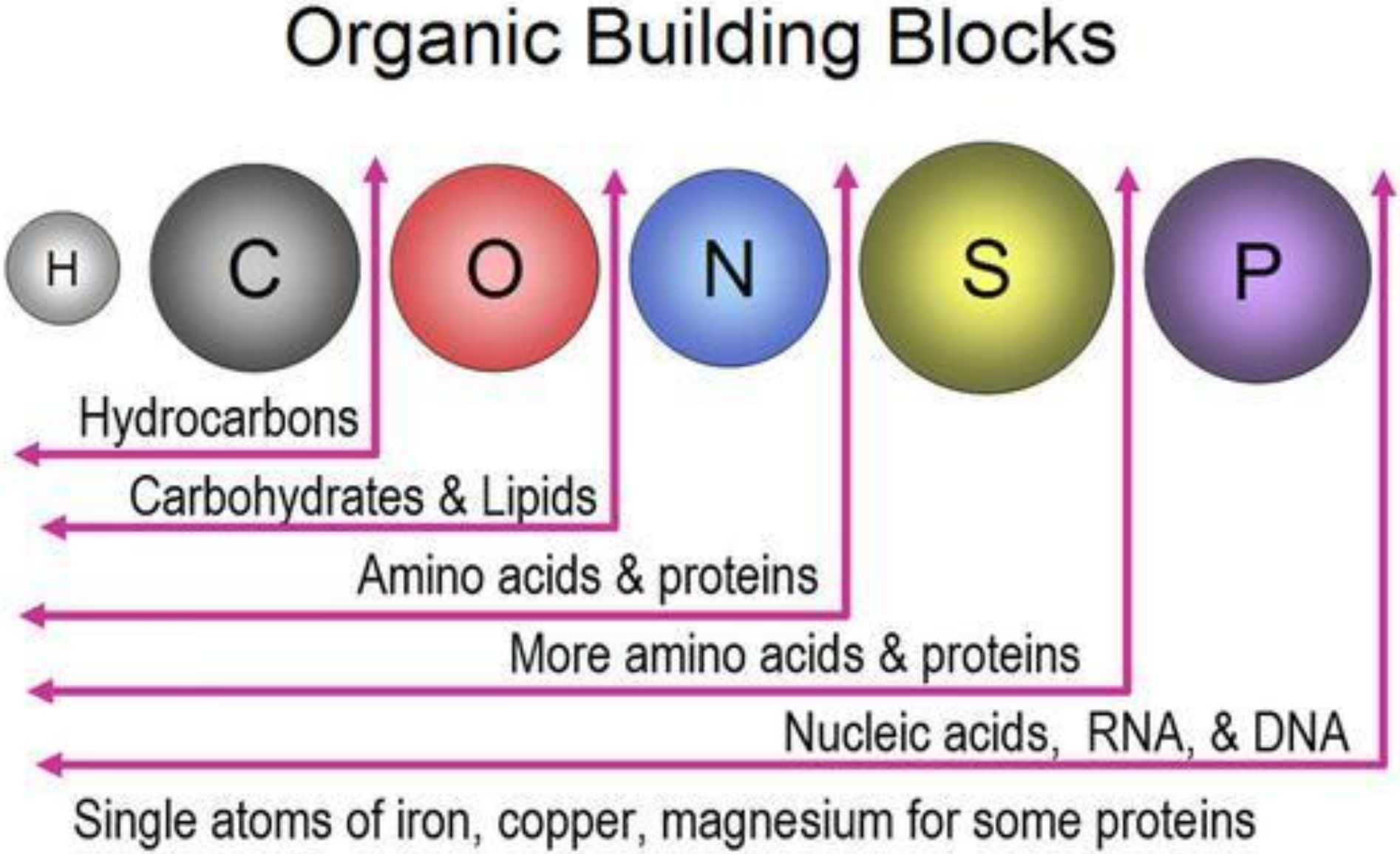
Lipids

Proteins

Nucleic Acids


THE HUMAN BODY
62%
Water
16%
Protein
Oxygen 65%
Carbon 18%
Hydrogen 9.5%
Nitrogen 3.2%
Calcium 1.5%
Phosphorus 1.2%
O
C
H
N
Ca
P
K
S
Na
Cl
Mg
Potassium 0.4%
Sulfur 0.2%
Sodium 0.2%
Chlorine 0.2%
Magnesium 0,1%
Other >1%
Minerals
6%
Fat
16%
1%
Carbohydrate
ELEMENTAL COMPOSITION

Biomolecules of life consists of six basic elements

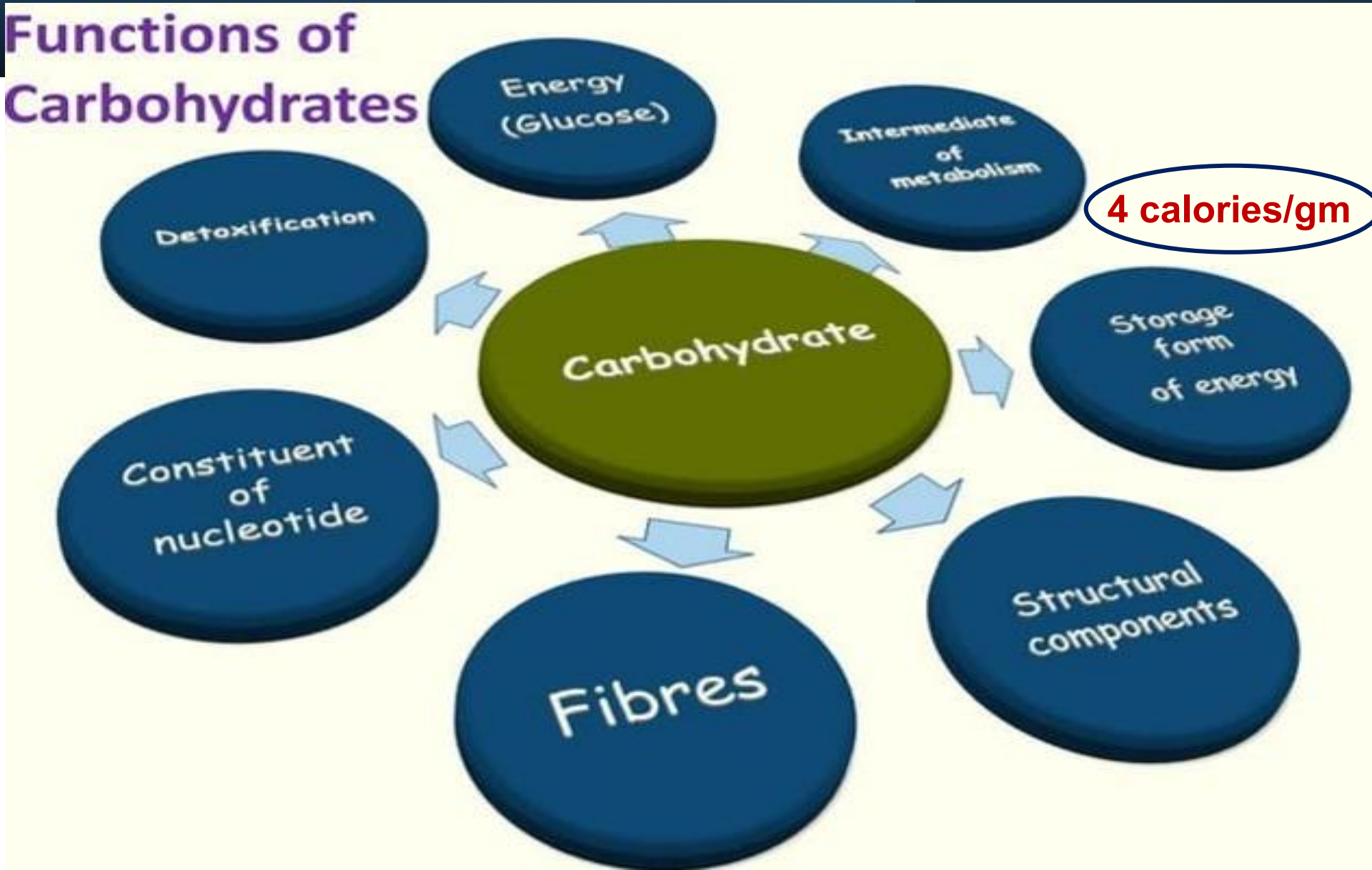


Biomolecule	Building block	Major functions
Protein	Amino acid	Basic structure and function of cell
DNA	Deoxyribonucleotide	Hereditary information
RNA	Ribonucleotide	Protein synthesis
Polysaccharide	Monosaccharide	Storage form of energy
Lipids	Fatty acids & glycerol	Storage form of energy to meet long term demands

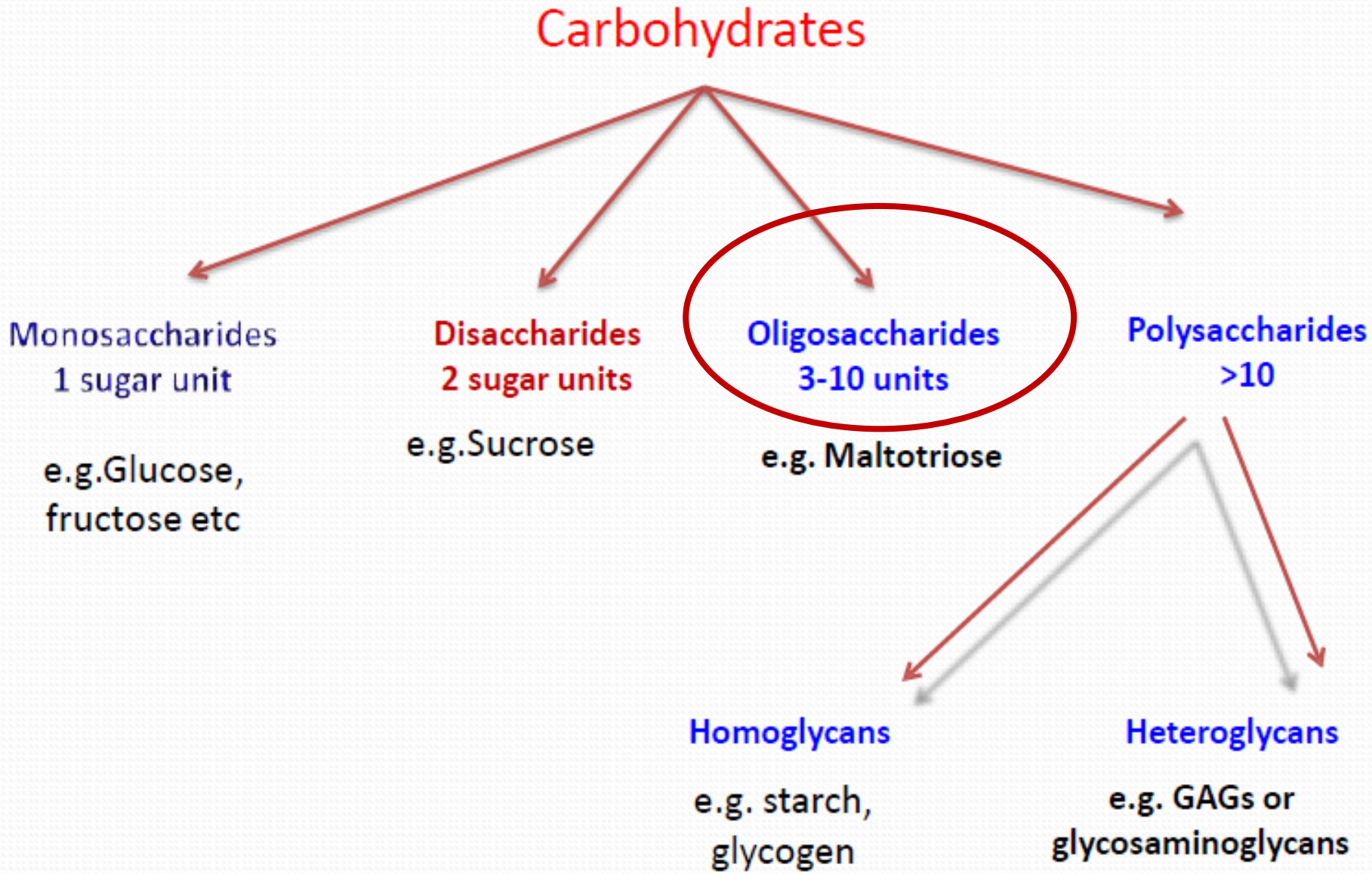
What are Carbohydrates?

- Most common organic molecule
- **Function:** Primary energy source our body needs
- **Elements present**: C, H, O (1:2:1 ratio) IMPORTANT!
- **Monomer (building block):** Monosaccharides (Glucose is most common)
- **Polymer:** Polysaccharides (starch, Glycogen, Cellulose, Chitin)
- **Examples**: Chocolate, Bread, Pasta, Fruits, Vegetables (ALL FROM PLANTS!!!)

Biological Importance of CARBOHYDRATES



Classification of Carbohydrates



CARBOHYDRATES

- One of the simplest carbohydrates is **GLUCOSE,** which is a type of **sugar** – single sugar – **MONOSACCHARIDE**
- Its chemical formula is: $C_6H_{12}O_6$
- One can find glucose in basically all living things.
- Glucose **dissolves** very easily in water therefore is very **soluble**.
- Glucose is present in all your cells and is the main substance which provides energy.



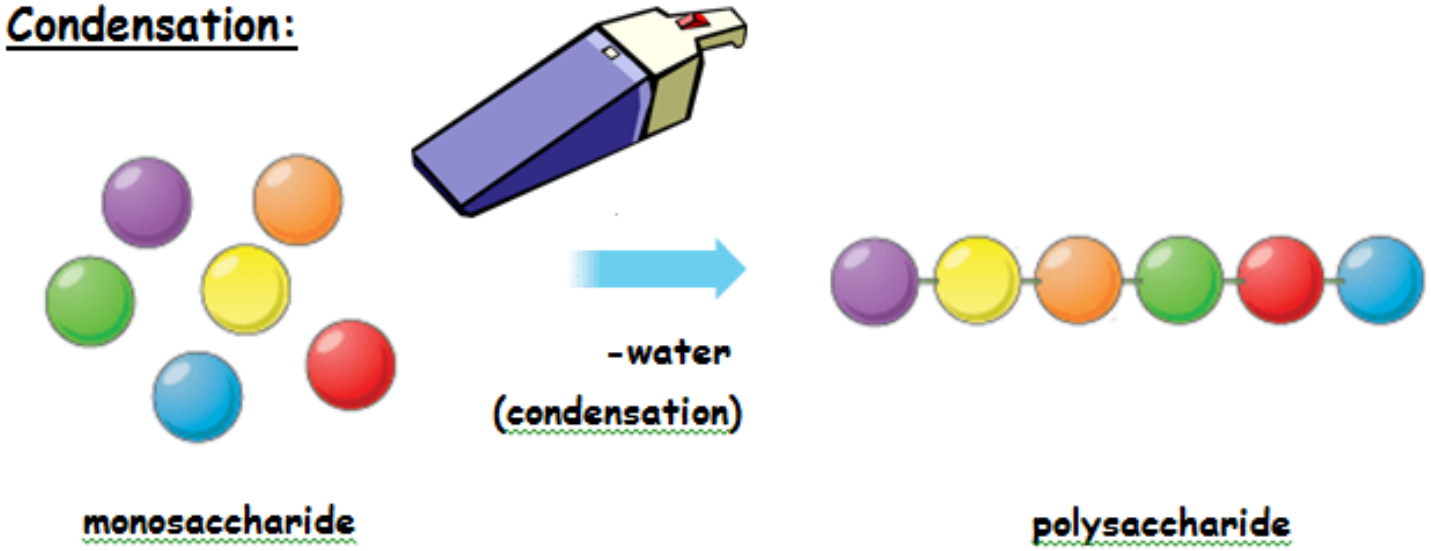
Other single-sugars are **fructose** and **galactose**.

CARBOHYDRATES

- Many molecules of glucose can join together to form a **multi-sugar** or **POLYSACCHARIDE**.
- One example of a **polysaccharide** is **glycogen**. **Glycogen** forms a food storage substance in many **animals**.
- **Starch** which is also a **polysaccharide** is the food storage substance usually used by **plants**.

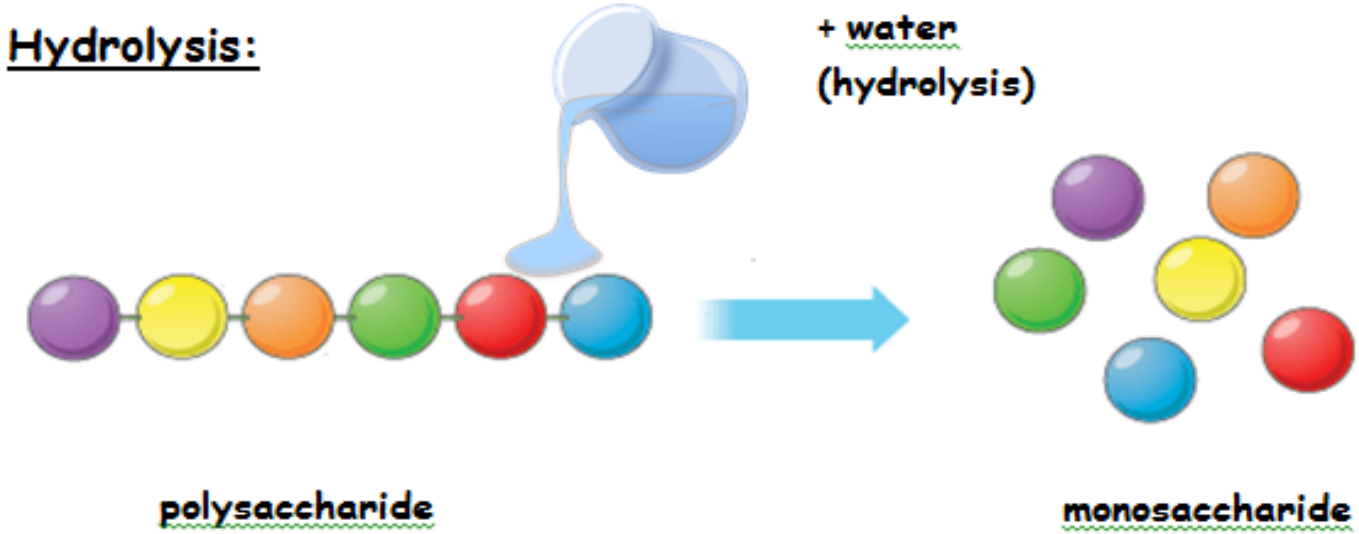
CARBOHYDRATES

- When glucose molecules join together **water** is **REMOVED** from them.
- This type of chemical reaction is called **CONDENSATION**



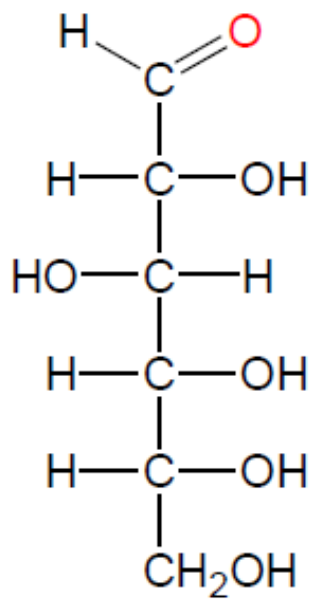
CARBOHYDRATES

- The opposite can happen for example starch broken down into glucose.
- For the glucose molecules to be separated from one another **water** has to be added.
- This type of reaction is called **HYDROLYSIS**.

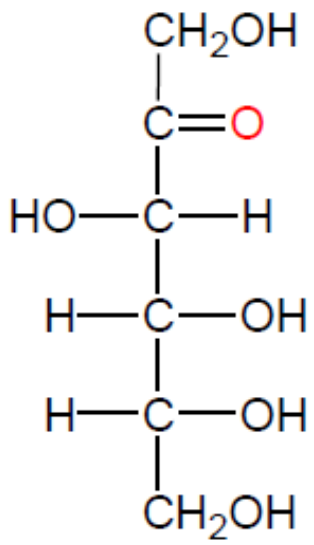


Monosaccharides:

They are the simplest carbohydrates that cannot be broken into smaller units on hydrolysis.



D-glucose



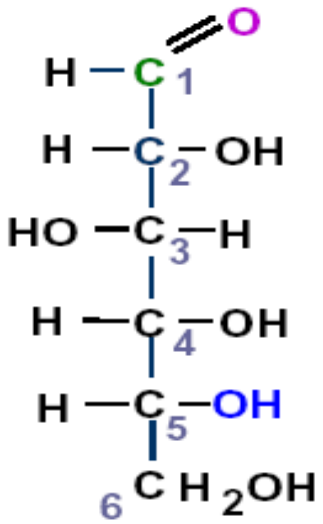
D-fructose

- Aldoses (e.g., glucose) have an aldehyde group at one end
- Ketoses (e.g., fructose) have a keto group, usually at C2.

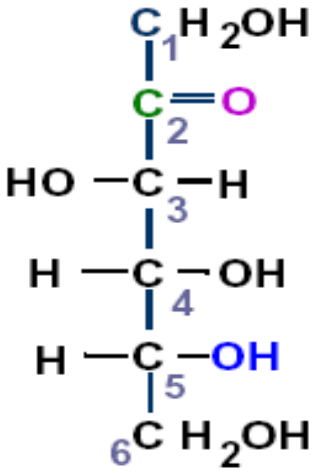
Monosaccharides

Aldoses (e.g., glucose) have an aldehyde group at one end.

Ketoses (e.g., fructose) have a keto group, usually at C2.



D-glucose
(an aldohexose)



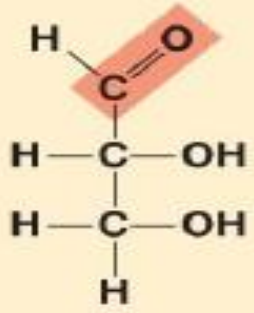
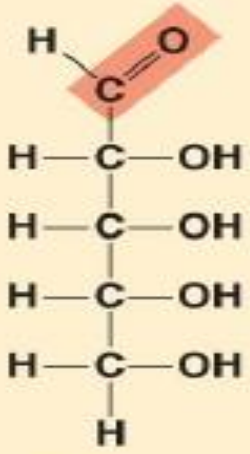
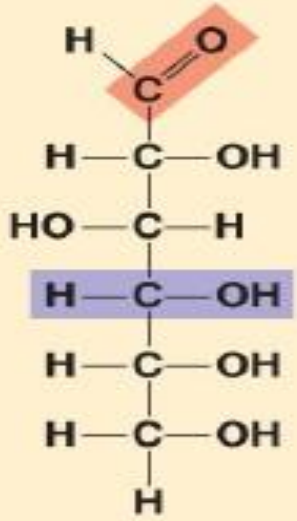
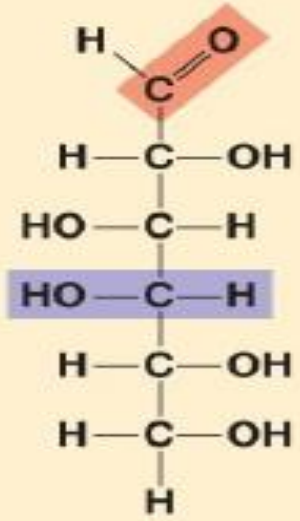
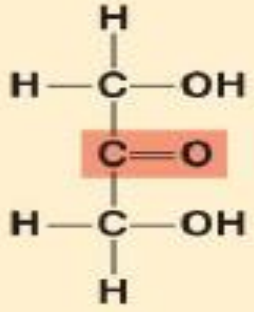
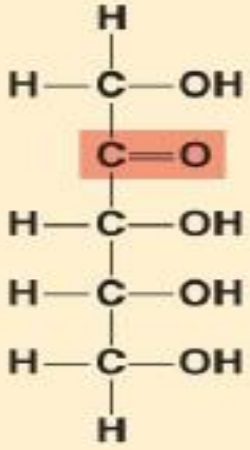
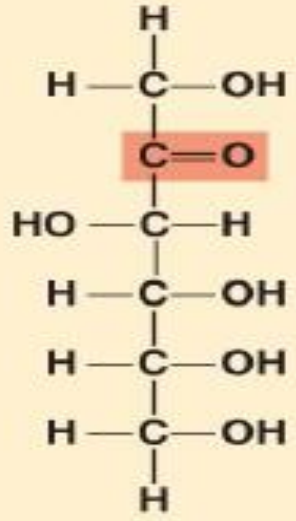
D-fructose
(an ketohexose)

Monosaccharide further classified on the basis of functional group and number of carbon atoms present in their structure

On the basis of no. of carbon atom

On the basis of functional group

No. of carbon atom	Generic name	ALDOSE	KETOSE
3	Trioses	Aldotriose e.g. glyceraldehyde	Ketotriose e.g. Dihydroxyacetone
4	Tetroses	Aldotetrose e.g. Erythrose	Ketotetrose e.g. Erythrulose
5	Pentoses	Aldopentoses e.g Arabinose, Xylose, Ribose	Ketopentoses e.g. Xylulose, Ribulose
6	Hexoses	Aldohexose e.g. Glucose, Galactose, Mannose	Ketohexose e.g. Fructose
7	Heptoses	Aldoheptose: Glucoheptose	Ketoheptose e.g Sedoheptulose

	Trioses ($C_3H_6O_3$)	Pentoses ($C_5H_{10}O_5$)	Hexoses ($C_6H_{12}O_6$)	
Aldoses	 Glyceraldehyde	 Ribose	 Glucose	 Galactose
Ketoses	 Dihydroxyacetone	 Ribulose	 Fructose	

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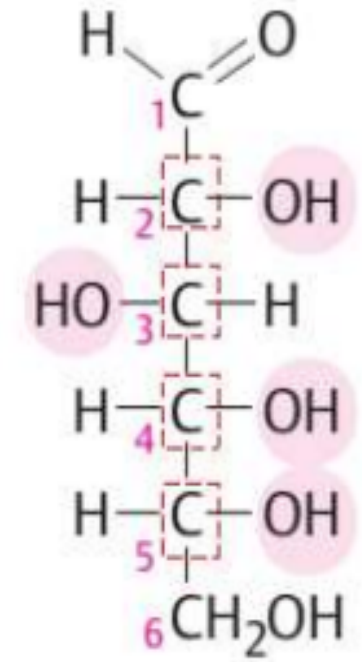
TYPES	EXAMPLE	IMPORTANCE
Trioses	Glyceraldehyde, Dihydroxyacetone	✔ Intermediates of glycolysis, ✔ Precursor of glycerol (for lipid synth)
Tetroses	D-Erythrose	✔ Intermediates of carbohydrate metabolism
Pentoses	D-Ribose	✔ Structural element of nucleic acid, RNA, co-enzymes.
Hexoses	D-Glucose	✔ Main sugar of the body.
	D-Fructose	✔ Converted to glucose & utilized by the body.
	D-Galactose	✔ Synthesized in mammary gland to make the lactose of milk.
	D-Mannose	✔ Constituent of glycoprotein, glycolipids

Isomerism in monosaccharides

- **Stereoisomerism** - Compound shaving same structural formula but they differ in spatial configuration. Example: Glucose has two isomers with respect to penultimate carbon atom. They are D-glucose and L-glucose.
- **Optical Activity** - It is the rotation of plane polarized light forming (+) glucose and (-) glucose.
- **Distereo isomers** - It the configurational changes with regard to C2, C3, or C4 in glucose. Example: Mannose, galactose.
- **Annomerism** - It is the spatial configuration with respect to the first carbon atom in aldoses and second carbon atom in ketoses.

Stereoisomerism

- Important characteristics of monosaccharide.
- Same structural formulae but differs in their spatial arrangement.
- A carbon is said to **assymetric or chiral** when its is attached to different atoms or groups.
- Possible number of isomers – 2^n where *n = number of chiral carbon atoms*.



Open-chained form of glucose

Chiral center

Enantiomers

- Non-superimposable *COMPLETE* mirror image
- A special type of isomerism.
- The two members of the pair are designated as **D** and **L** forms.
- Majority of the sugars in humans are D-sugars.

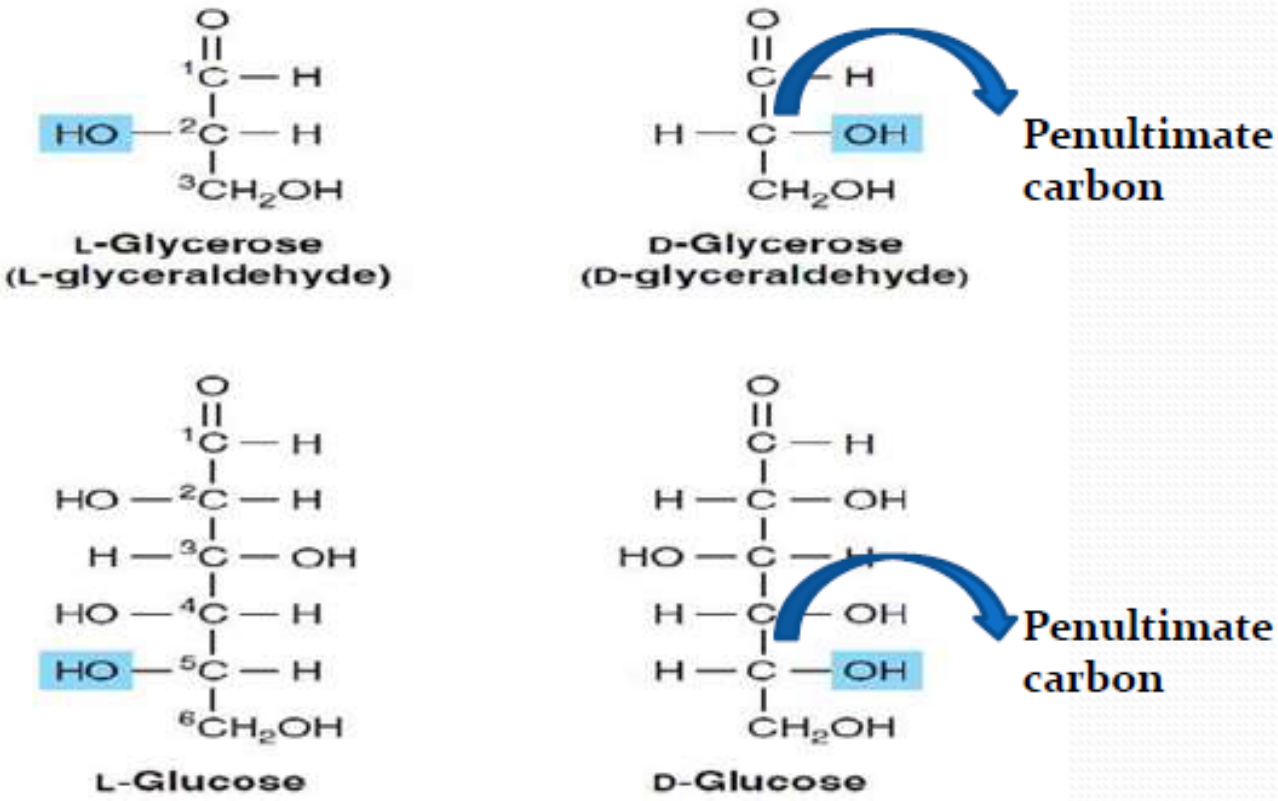


D- and L- isomers

D and L Isomers are mirror images of each other.

The *spatial orientation of H & OH groups on the C- atom (C_5 for glucose), adjacent to the terminal primary alcohol carbon determines whether the sugar is D or L Isomer.*

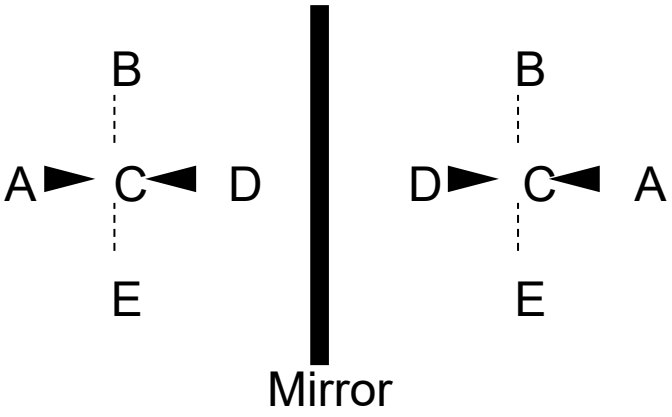
If the OH group is on the right side, the sugar is of D- Isomer. If the OH group is on the Left side, the sugar is of L- Isomer. Mammalian tissues have D- sugars.

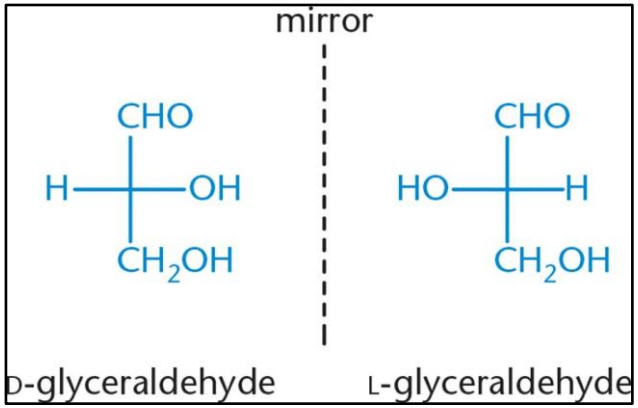


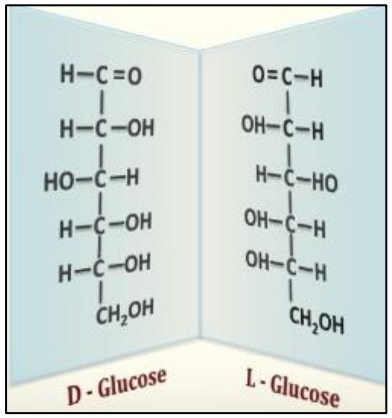
Optical Isomerism

- Optical activity is characteristic principle of asymetric carbon atom.
- Polarized light when passed through solution
  - turns to left – **Levorotatory (–)**
  - and if it turns to right – **Dextrorotatory (+).**
- So carbohydrate can be either D(+), D(–), L(+), L(–).
- **Racemic mixture** : when D and L isomers are present in equal concentration which doesn't shows any optical activity.

Enantiomers- Mirror images and non superimposable on each other







Ex: D-Glucose and L-Glucose

Distereoisomers

- **Non mirror images and non superimposable**

Ex: Glucose and Mannose

- **Dextrorotatory (d)** which is labelled (+) rotate a plane of polarized light in a clockwise direction.
- **Levorotatory (l)** which is labelled (-) rotate the light in a counterclockwise direction.

Epimers

- Epimers are sugars that differ in configuration at ONLY one specific carbon atom.
- With the exception of the carbonyl carbon
- Examples of epimers :
  - glucose & galactose (epimeric at C4)
  - glucose & mannose (epimeric at C2)

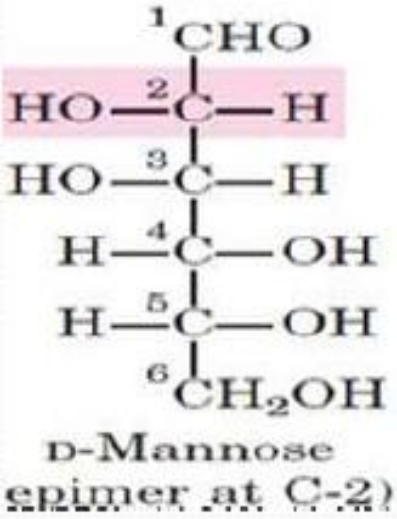
Epimers

If two monosaccharides differ from each other in their configuration around a single specific carbon atom, they are referred as epimers to each other.

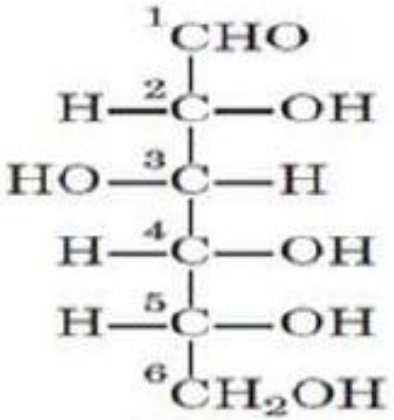
Glucose & galactose are C4-epimers.

Glucose & mannose are C2-epimers

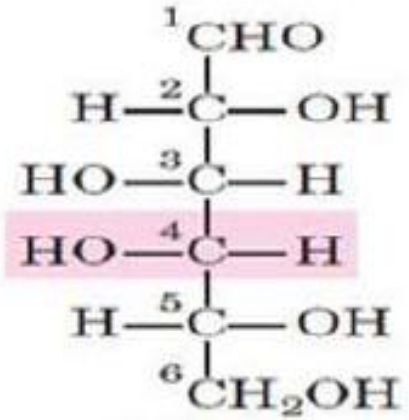
Inter-conversion of epimers is known as epimerization, epimerase catalyzes this reaction.



D-Mannose
(epimer at C-2)



D-Glucose



D-Galactose
(epimer at C-4)

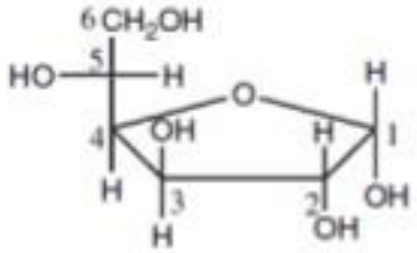
Cyclization of Monosaccharides

- Less than 1% of monosaccharides exist in the open chain form in solution.
- Predominantly found in ring form or cyclic structure.
- Involving reaction of C-5 OH group with the C-1 aldehyde group or C-2 of keto group.

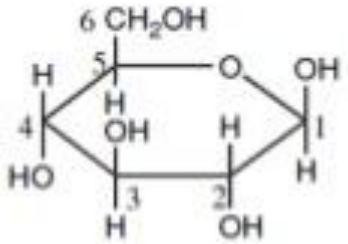
Haworth Projection (Cyclic Form)



W. N. Haworth (Nobel Prize, 1937)



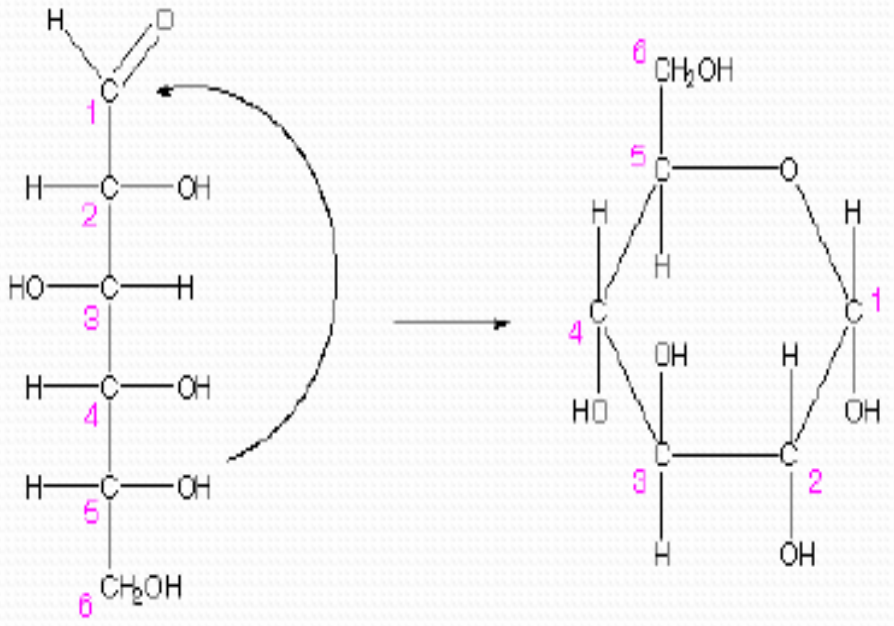
42, α-D-glucofuranose



41, β-D-glucopyranose

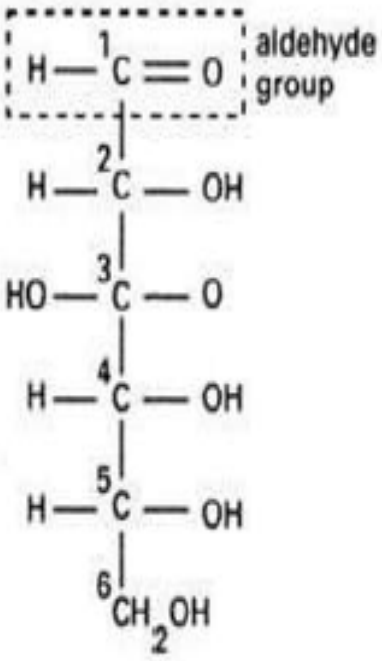
- Cyclization via intramolecular hemiacetal (hemiketal) formation
- C-1 becomes chiral upon cyclization - anomeric carbon
- Anomeric C contains -OH group which may be **α** or **β** (mutarotation **α** ⇔ **β**)

- Six membered ring structures are called **Pyranoses .**
- Five membered ring structures are called **Furanoses .**

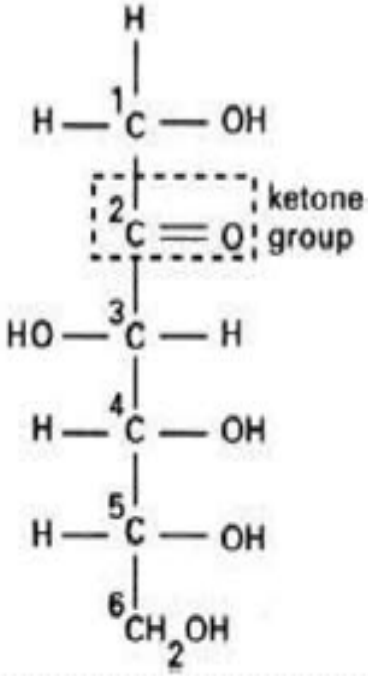


Straight chain form

Glucose (an aldohexose)

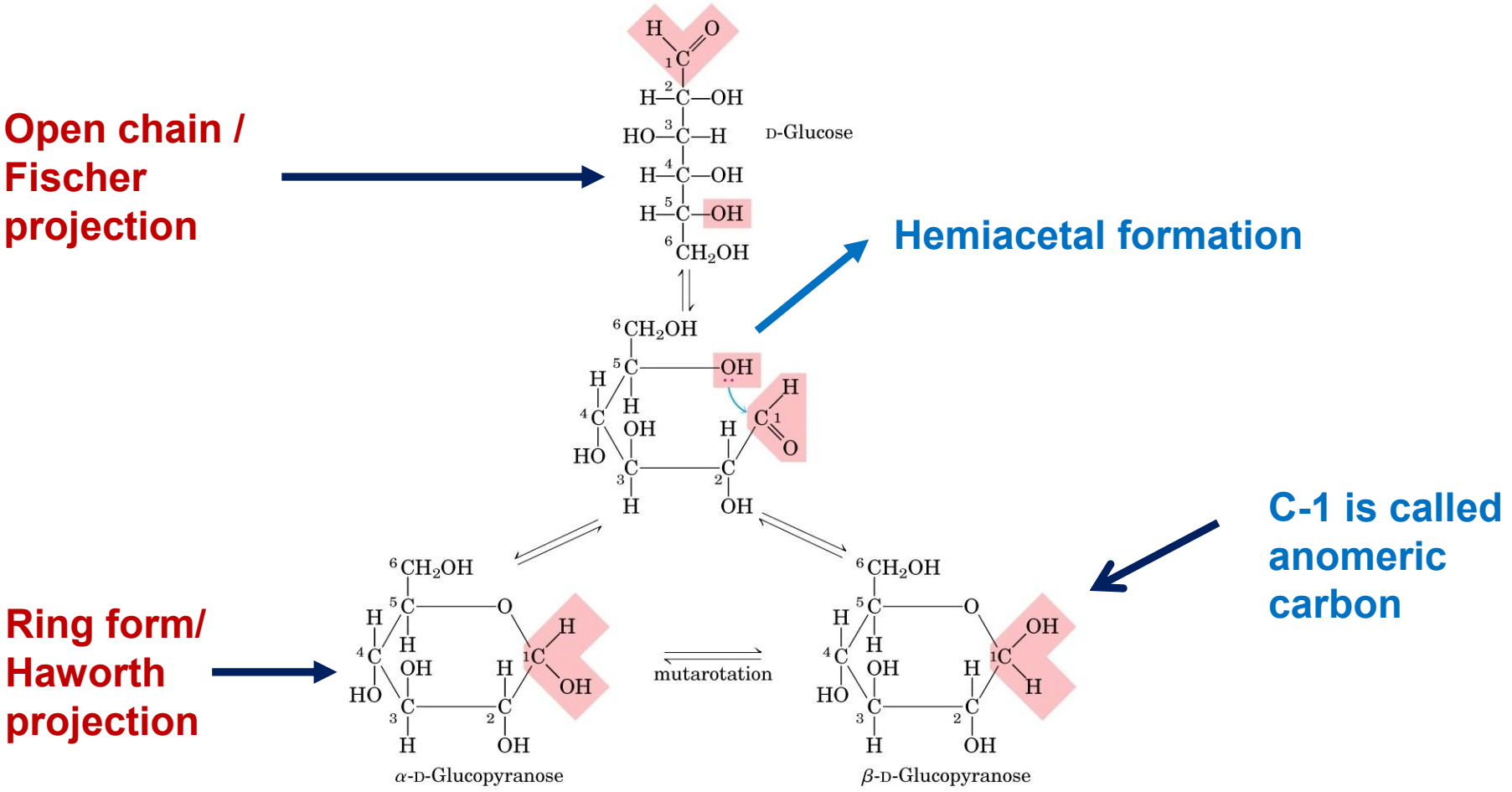


Fructose (a ketohexose)



Glucose

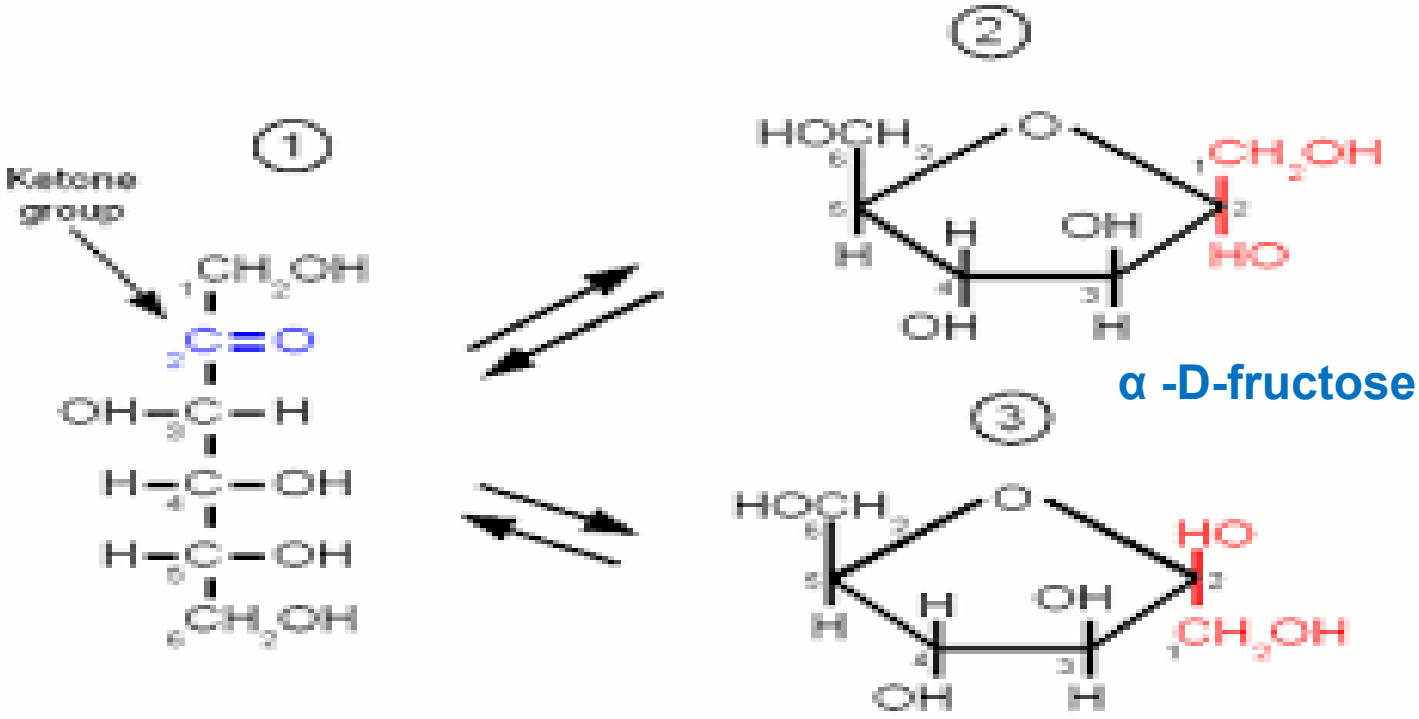
- **C-1 and C-5-intramolecular hemiacetal formation give rise to a pyranose ring**



Pyranose ring

Fructose

- **C-2 and C-5-intramolecular hemiketal formation in fructose forms furanose ring**



C-2 is called anomeric carbon

Furanose ring

Anomerism – Mutarotation

Anomers have same composition but differ in the orientation of groups around anomeric carbon atom.

Anomeric carbon is a hemiacetal or carbonyl carbon atom, e.g. 1st carbon atom in glucose is anomeric carbon atom.
Carbonyl carbon atom becomes asymmetric because of ring structures of monosaccharides in solution thus anomers are encountered in cyclic structures of monosaccaharides.

The alpha & beta cyclic forms of D-glucose are known as Anomers. They differ from each other in the configuration only around C1 known as anomeric carbon.

In case of alpha anomer, the OH group held by anomeric carbon is on the opposite side of the group CH_2OH of sugar ring

Anomers are expressed as α and β forms.

In α form "OH" group is below the plane (OH group is oriented away from the oxygen atom)

In β form "OH" group is above the plane (OH group is oriented towards the oxygen atom)

Mutarotation:

When D-glucose is crystallized at room temperature and a fresh solution is prepared, its specific rotation of polarized light is 112°; but after 12- 18 hrs it changes to +52.5 °

Mutarotation is defined as the change in the specific optical rotation representing the interconversion of α and β forms of D-glucose to an equilibrium mixture.

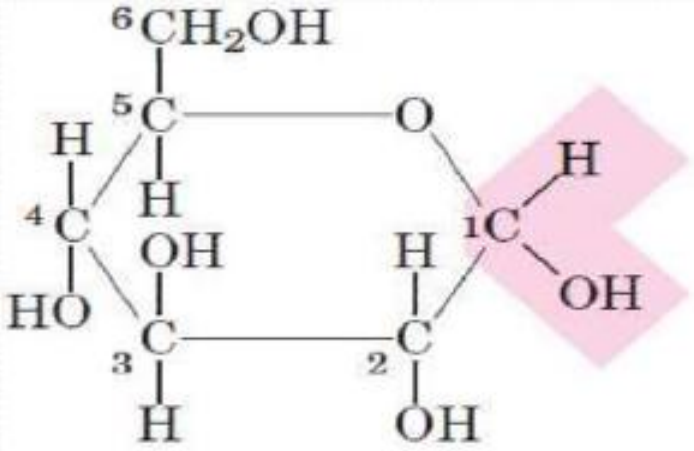
- α-D- glucose is less stable as compared to β- D- glucose.
- If we will dissolve α-D- glucose in water then some part will convert into β- D-glucose and achieve the equilibrium I the solution and if we will take β- D-glucose then reverse condition will occur.

Specific rotation

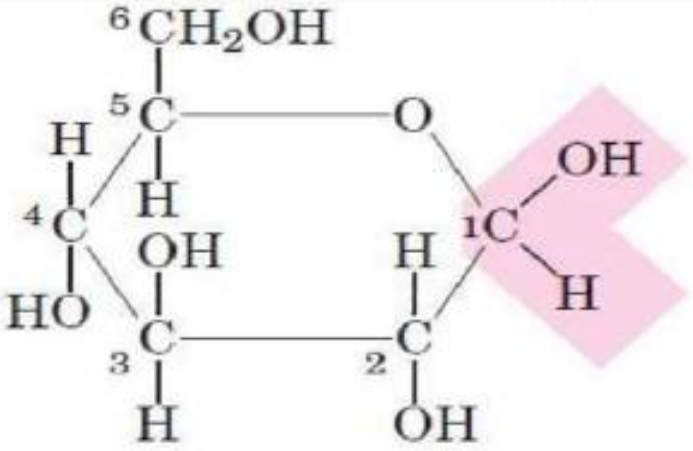
- α-D- glucose ⟷ Equilibrium mixture ⟷ β- D- glucose

  +112.2° +52.4° +18.7°

- Due to more stability β- D- glucose will be present in higher percentage.

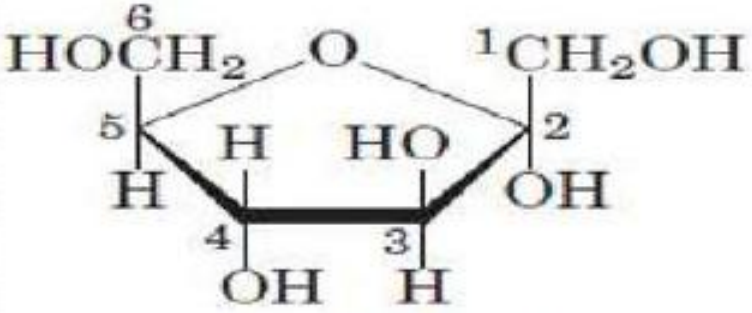


α-D-Glucopyranose

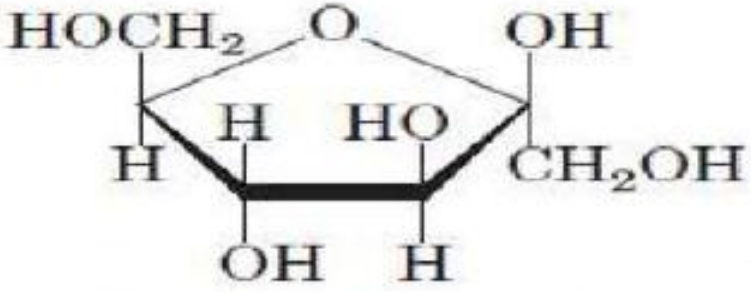


β-D-Glucopyranose

Glucose and Fructose has two anomers α and β



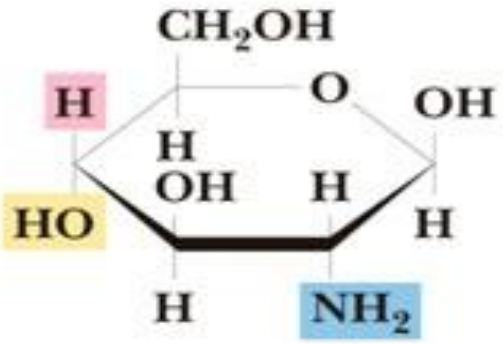
α-D-Fructofuranose



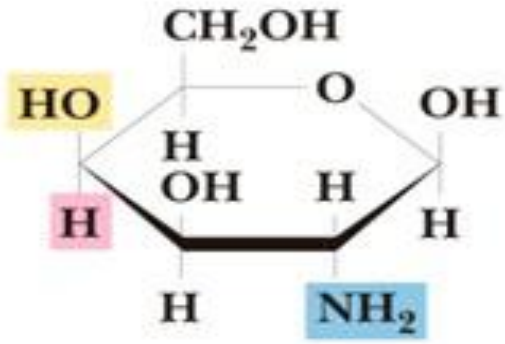
β-D-Fructofuranose

Other monosaccharide derivatives

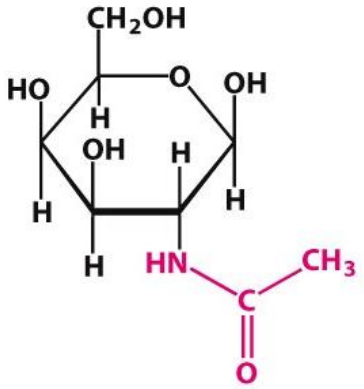
- **Amino sugars** found in animal cells



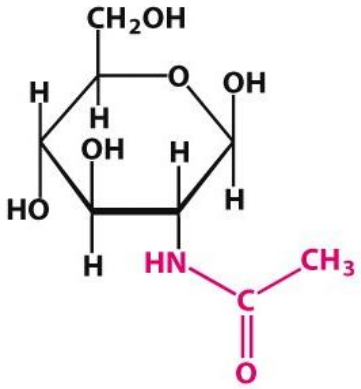
β-D-Glucosamine



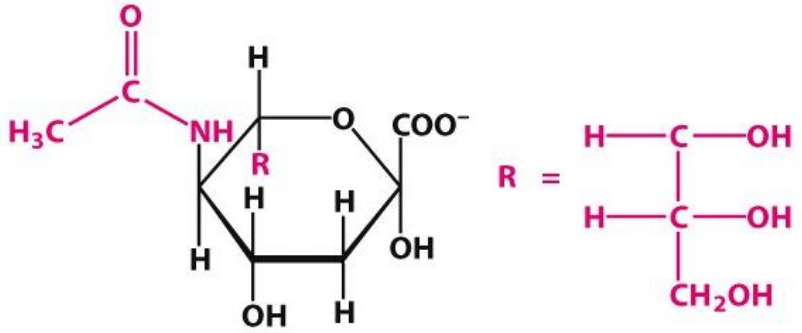
β-D-Galactosamine



β-D-Acetylgalactosamine (GalNAc)



β-D-Acetylglucosamine (GlcNAc)



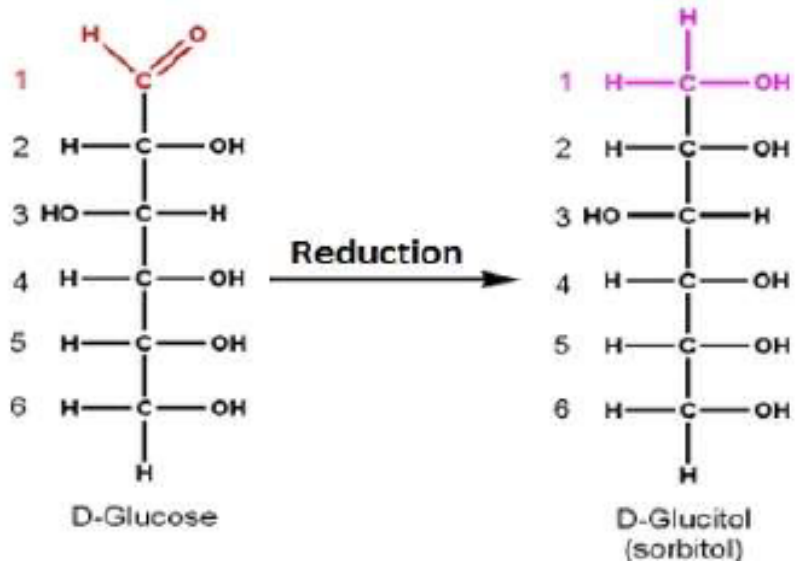
Sialic acid (Sia)
(*N*-Acetylneuraminate)

PROPERTIES OF MONOSACCHARIDES

1. **Physical properties:**
   Monosaccharides are colourless, crystalline compounds, readily soluble in water and sweet in taste. Their solutions are optically active and exhibit mutarotation.

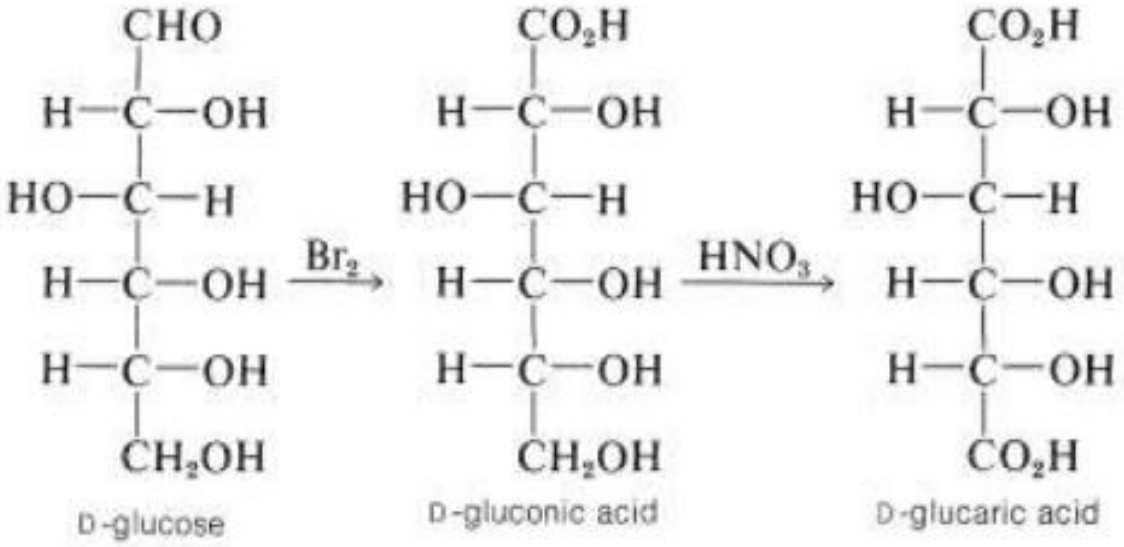
2. **Chemical properties:**
   a) **Reduction:** When treated with reducing agents such as sodium amalgam, hydrogen can reduce sugars. Aldose yields corresponding alcohol. Ketoses form two alcohols because of appearance of new asymmetric carbon in this process.

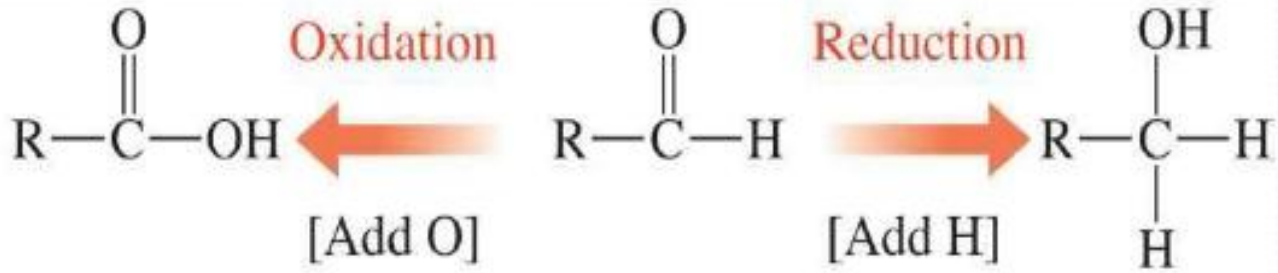
D-Glucose → D-Sorbitol
D-Fructose → D-Sorbitol+D-Mannitol



b) Oxidation: upon oxidation aldose sugars like glucose yields carboxylic acid, *the aldehyde group is oxidized to carboxyl group to produce respective acids.*

Glucose → Gluconic acid
Mannose → Mannonic acid
Galactose → Galactonic acid



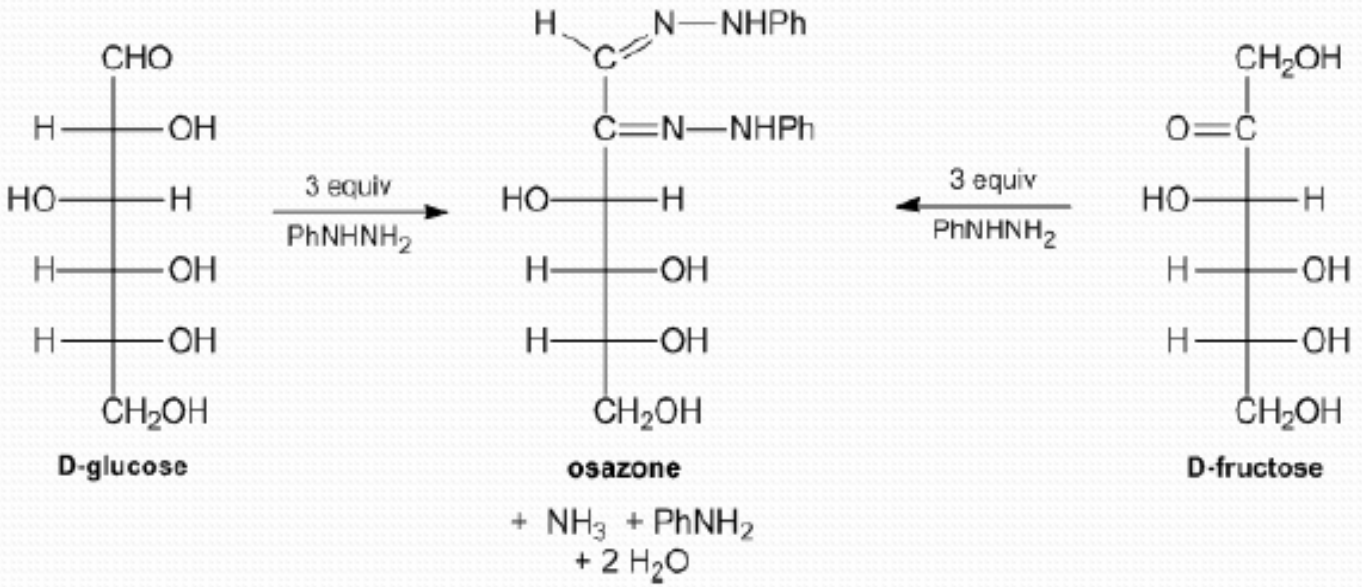


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c) Formation of esters: Hydroxyl groups of sugars can be esterified to form acetates, propionates, benzoates, etc. Sugar phosphates are of great biological importance. Metabolism of sugars inside the body starts with phosphorylation.

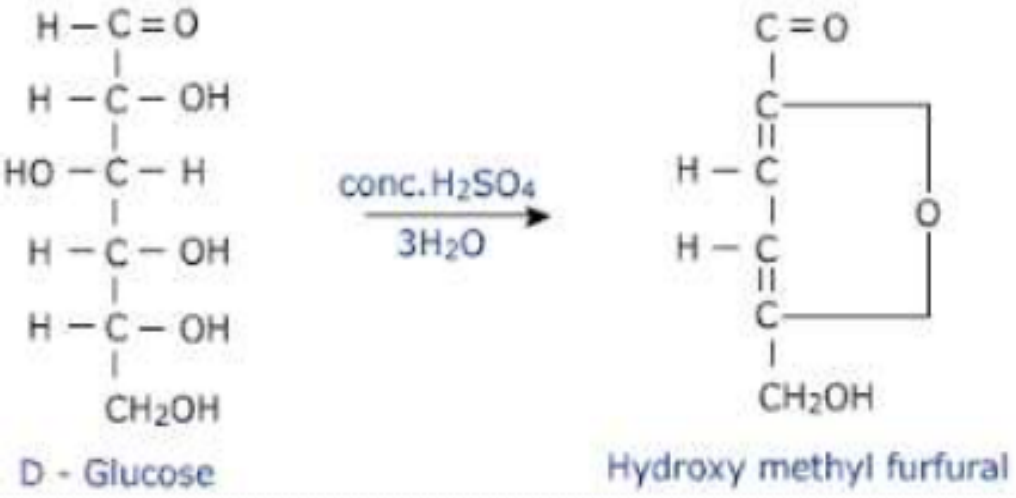
e.g Glucose 6-Po_4

d) Formation of osazone: this test is given by reducing sugar like, glucose, fructose, lactose and maltose. In this test phenyl hydrazine is reduced to phenyl hydrazone by sugar solution. Phenyl hydrazone when heated with more amount of phenyl hydrazine forms yellow crystals of osazone.



e) Furfural formation/Dehydration: Monosaccharides when treated with concentrated H_2SO_4 undergoes dehydration with the removal of 3 molecules of water. Hexoses give hydroxymethyl furfural and pentoses give furfural. Furfurals condense with phenolic compounds to give various colors.

E.g. Molisch's test: General test for carbohydrates (H_2SO_4 and α-naphthol). Rapid Furfural and Seliwanoff's test: Tests for presence of keto group



DISACCHARIDES

A disaccharide consists of 2 monosaccharide units (similar or dissimilar) held together by a glycosidic bond

They are crysatalline, water soluble and sweet to taste.

MALTOSE : - is also called as **malt sugar**.
- made up of **2 glucose molecules**

LACTOSE : - is also called **as milk sugar** as it is found naturally in milk
- made up of **glucose and galactose**
- souring of milk is due to conversion of lactose to lactic acid

SUCROSE : -is also called as **cane sugar.** It is the sugar found in sugar cane and sugar beet
- most abundant among naturally occuring sugars.
- Important source of Dietary carbohydrates
- made up of **glucose and fructose**

Glycosidic bonds

Linkage formed between two-sugar units involving –OH groups or a sugar unit with –OH group and N-atom containing compound such as amino acid asparangine, and glutamine; purine and pyrimidine bases

O-glycosidic bond: formed between sugar units or monosaccharides to form disaccharides, oligosaccharides or polysaccharides. Also found between sugar unit and –OH containing amino acid

They are either alpha-glycosidic bond or Beta glycosidic bond in disaccharides

Glycosidic bonds

Type is based on the position of the C-1 OH

α glycosidic bond
- linkage between a C-1 α OH and a C-4 OH

β glycosidic bond
- linkage between a C-1 β OH and a C-4 OH



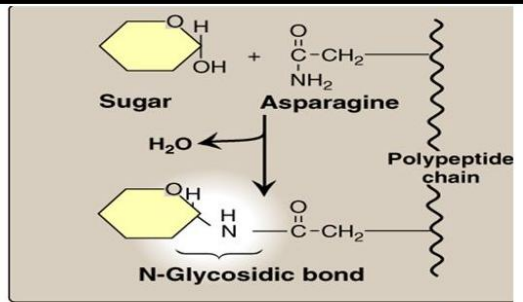
C-4 end can be either up or down depending on the orientation of the monosaccharide.

41

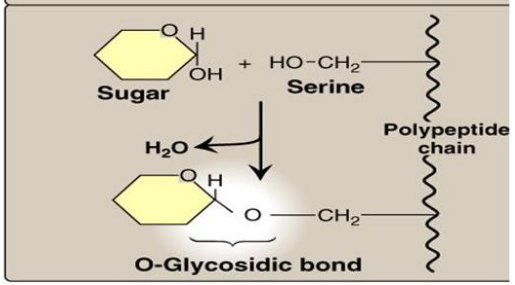
N-glycosidic bond: formed between a sugar unit with –OH group and N-atom containing compound **such as amino acid asparangine, and glutamine; purine and pyrimidine bases**.

Found in heteropolysaccharide such as proteoglycans, glycoproteins, and nucleic acids (DNA and RNA that has purine and pyrimidine bases

N-Glycosidic (C-N-C link)

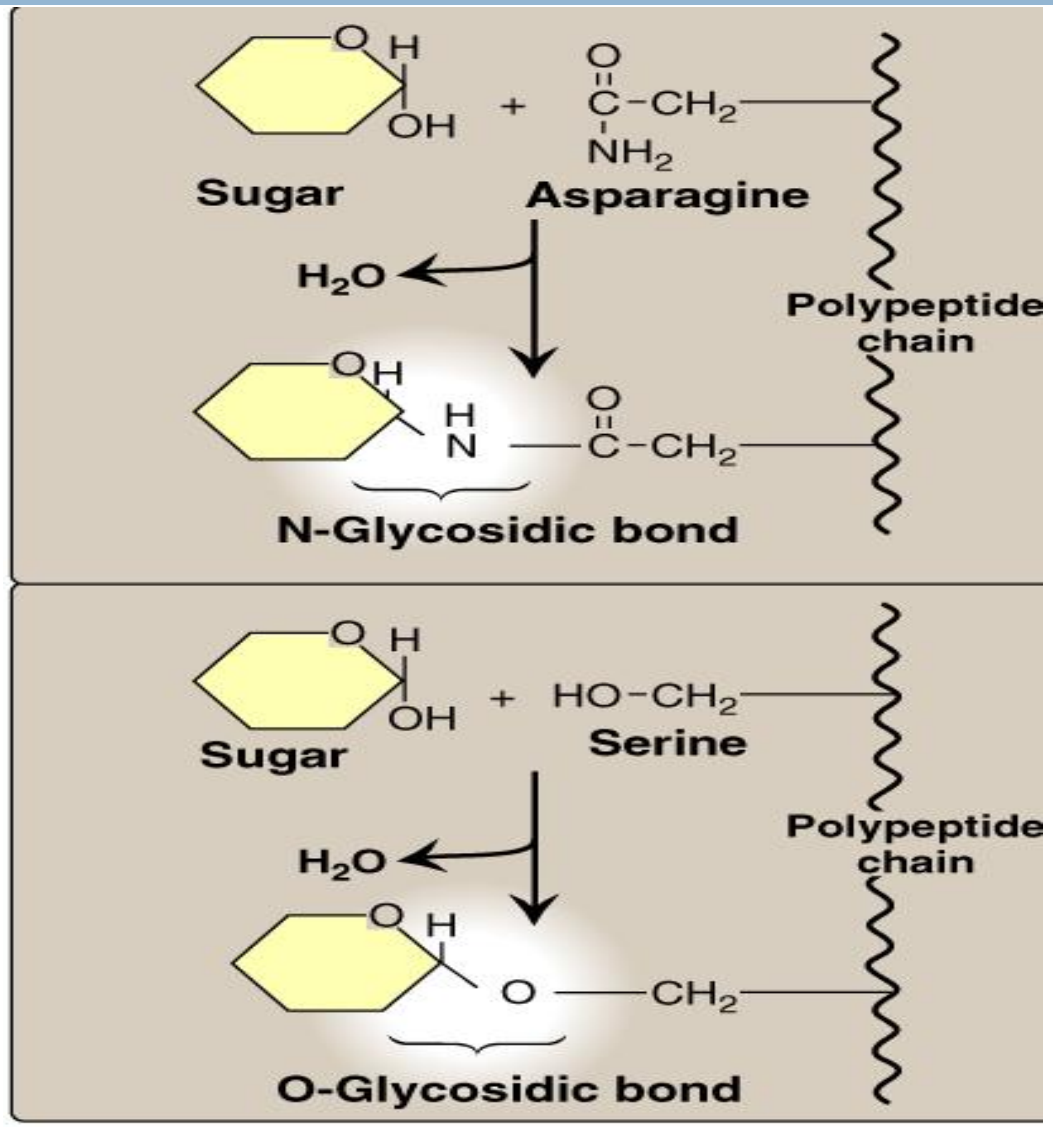


O-Glycosidic (C-O-C link)



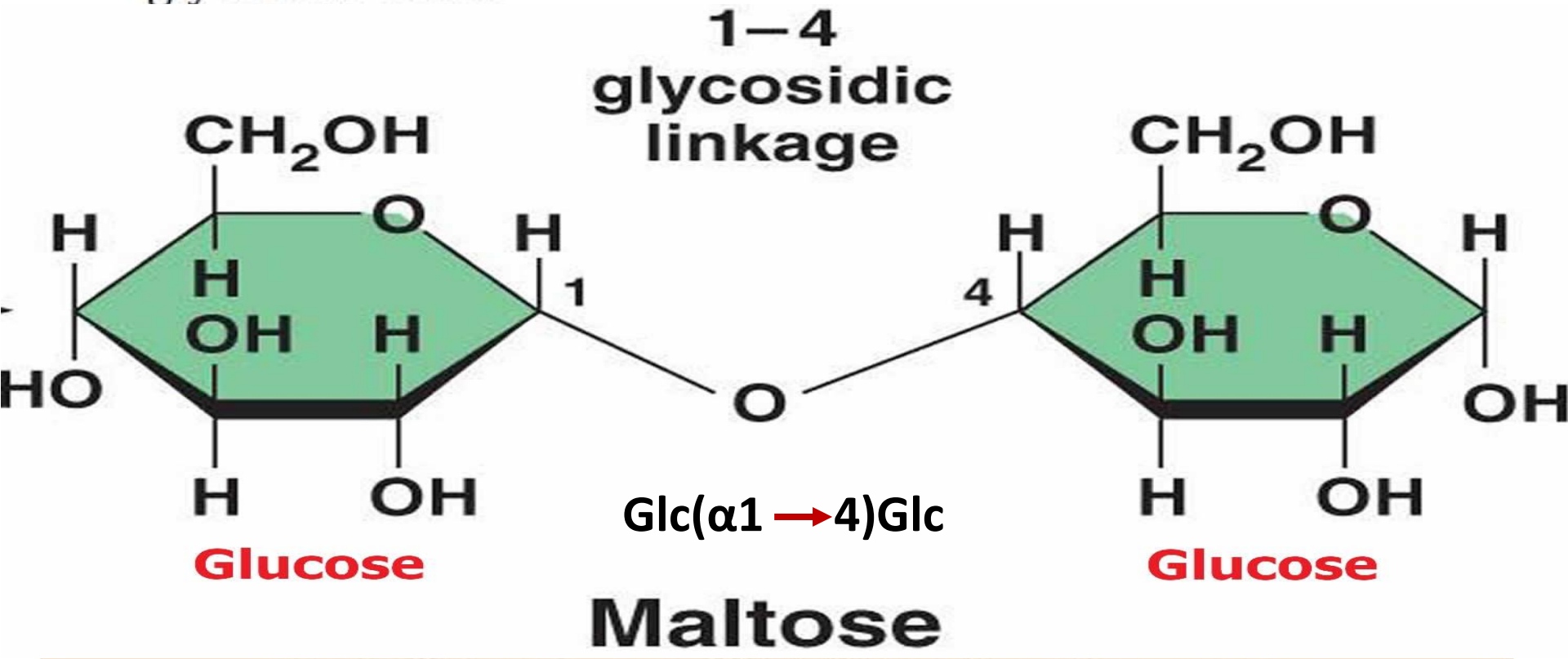
Glycosidic Bonds

- N-Glycosidic
- O-Glycosidic



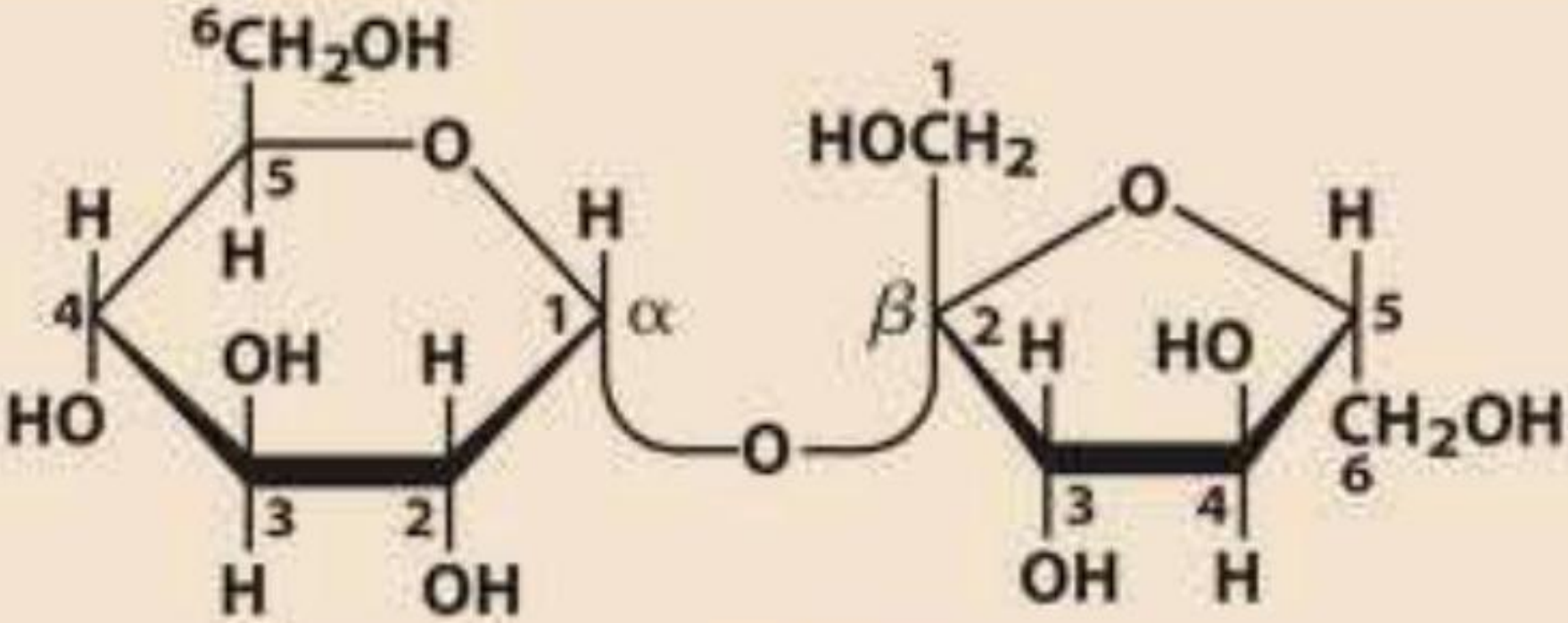
MALTOSE (Malt sugar)

- Maltose is formed by joining of 2 glucose units by α-(1,4) glycosidic bond.



α -D-glucopyranosyl- (1-4) -D -glucopyranose

SUCROSE (TABLE SUGAR)

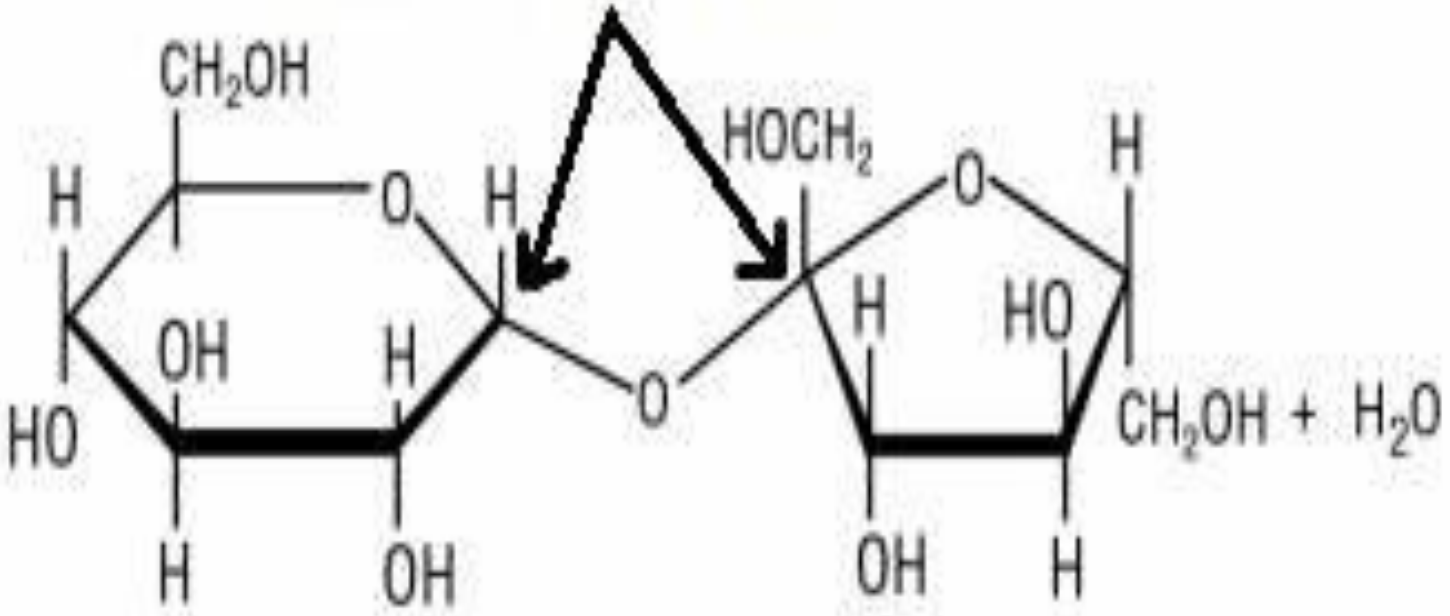


Sucrose

α-D-glucopyranosyl β-D-fructofuranoside

Glc(α1↔2β)Fru

Both anomeric carbons are involved in the O-glycosidic bond



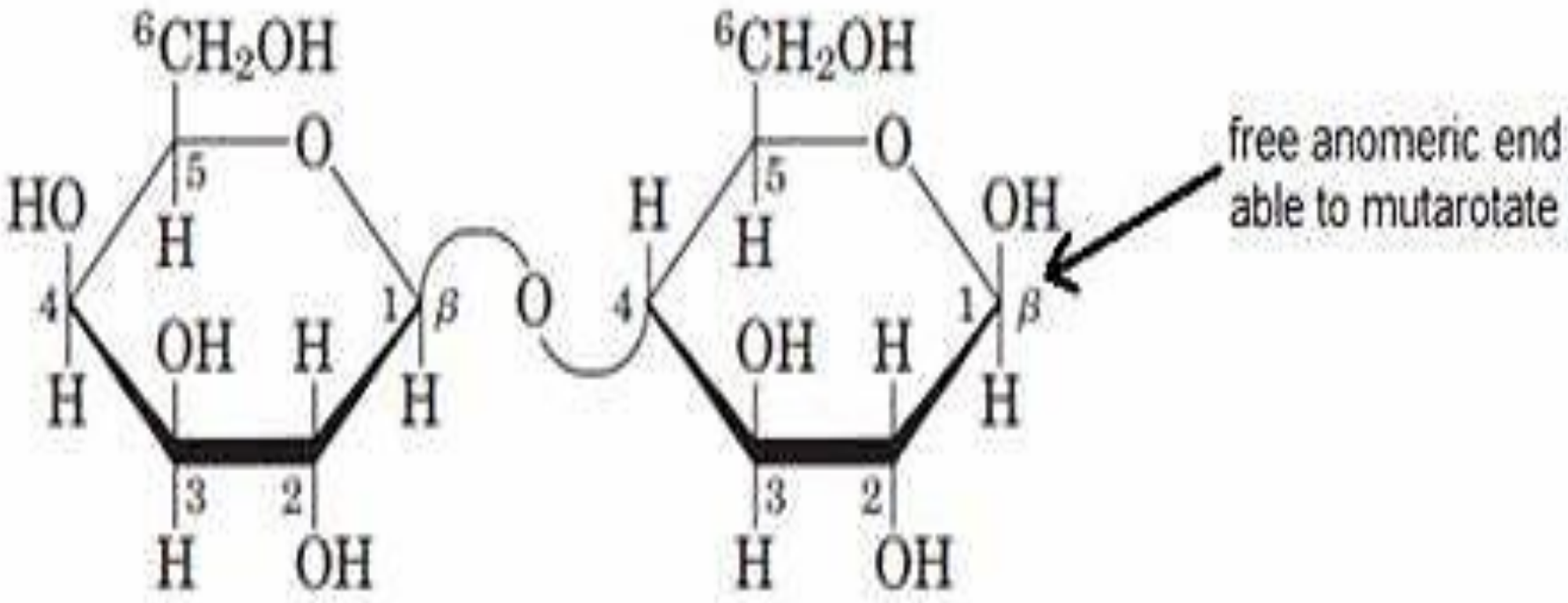
Sucrose

- Sucrose has no free reactive group because the anomeric carbons of both monosaccharides units are involved in the glycosidic bond.

- Thus, sucrose neither shows reducing properties nor mutarotation characters.

- Hydrolytic product of Sucrose is called invert sugar because the optical activity of sucrose (dextrorotatory) is inverted after hydrolysis [by an acid or an enzyme (invertase or sucrase)] into equimolar mixture of its two components glucose (+52.5) and fructose (-92.5) and the optical activity of the mixture becomes levorotatory.

LACTOSE (MILK SUGAR)



Lactose (β form)

β-D-galactopyranosyl-(1→4)-β-D-glucopyranose

Gal(β1→4)Glc

- Lactose is the only sugar of milk -

  Synthesized by mammary glands during lactation.

  Milk contains the α and β-anomers in a 2:3 ratio.

  β -lactose is sweeter and more soluble than ordinary α – lactose.

  used in infant formulations, medium for penicillin production and as a diluent in pharmaceuticals.

Reducing Sugars

- **Examples:**

  Monosaccharides

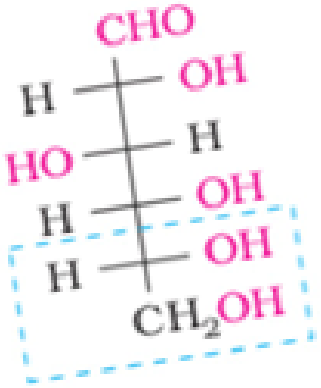
  Maltose and Lactose

  Sucrose is non-reducing, Why?

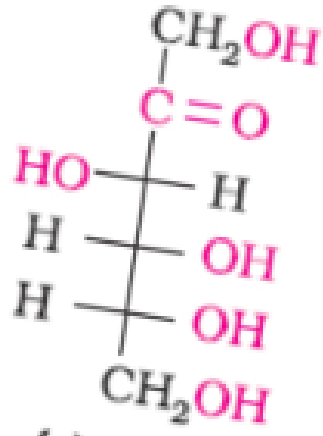
Reducing and Non- Reducing sugars

I. All those carbohydrates **which reduce Fehling's solution and Tollens' reagent** are referred to as reducing sugars.

II. **All monosaccharides whether aldose or ketose are reducing sugars.**

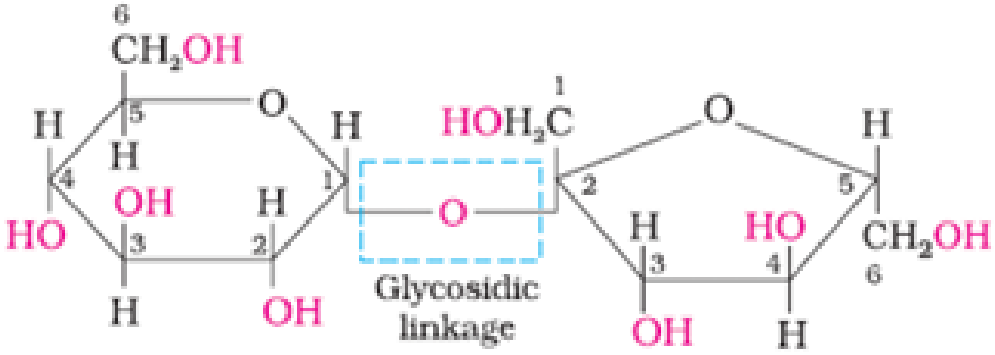


D–(+) – Glucose



D – (–) – Fructose

II. In disaccharides, if the reducing groups of monosaccharides i.e., aldehydic or ketonic groups are bonded, these are non-reducing sugars e.g. sucrose.

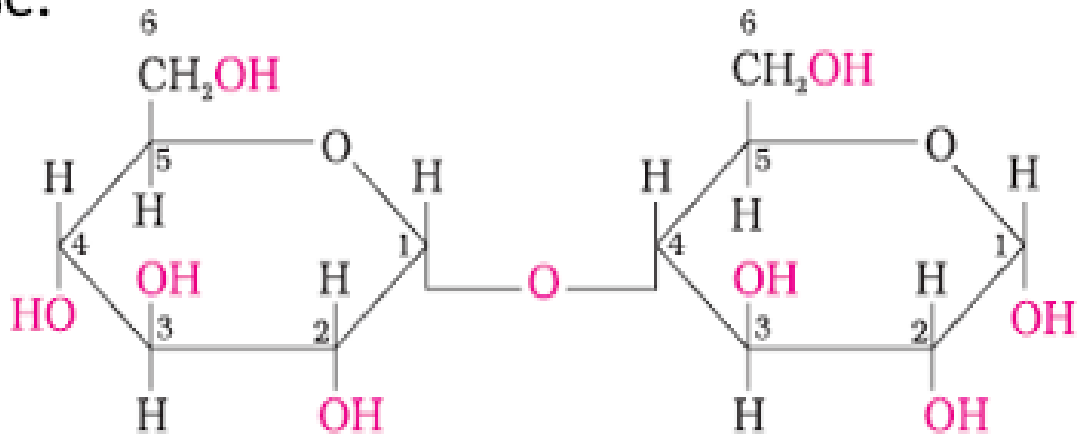


α – D – Glucose

β – D – Fructose

Sucrose

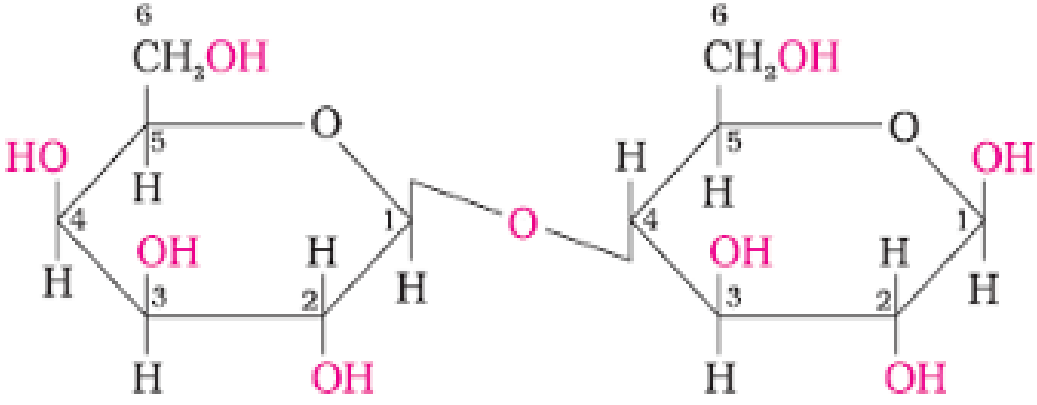
III. On the other hand, sugars in which these functional groups are free, are called reducing sugars, for example, maltose and lactose.



(I) α – D – Glucose

(II) α – D – Glucose

Maltose



β – D – Galactose

β – D – Glucose

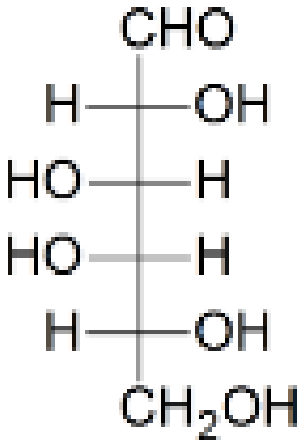
Lactose

OLIGOSACCHARIDES

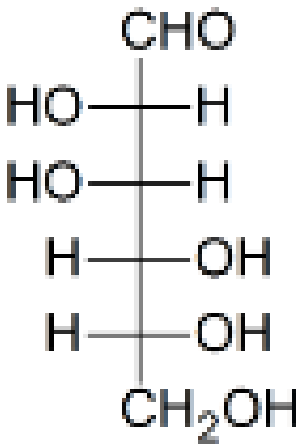
- Trisaccharide: Raffinose (glucose + galactose + fructose)
- Tetrasaccharide: Stachyose (2 galactose + glucose + fructose)
- Pentasaccharide: Verbascose (3 galactose + glucose + fructose)
- Hexasaccharide: Ajugose (4 galactose + glucose + fructose)

Sample questions

(01) Study the following Fischer projections to answer the questions below.



galactose



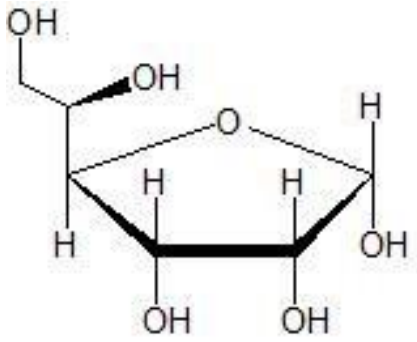
mannose

I. Is galactose a D-sugar or an L-sugar?
II. Is mannose a D-sugar or an L-sugar?
III. Choose either one and sketch it as it would appear if it was an L-sugar.
IV. Are these two carbohydrates enantiomers? If not, in how many places do they differ?
V. What is the term to describe the relationship between galactose and mannose?

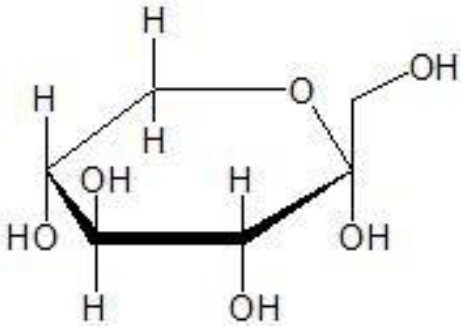
(02) Which carbon is considered the anomeric carbon? How do you distinguish between the alpha and beta types of anomers?

(03) Study these Haworth projections to answer the following for each of them.

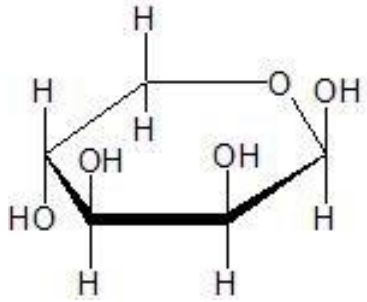
A)



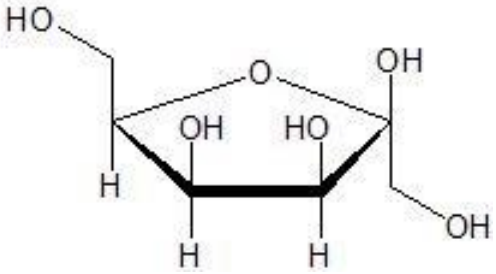
B)



C)

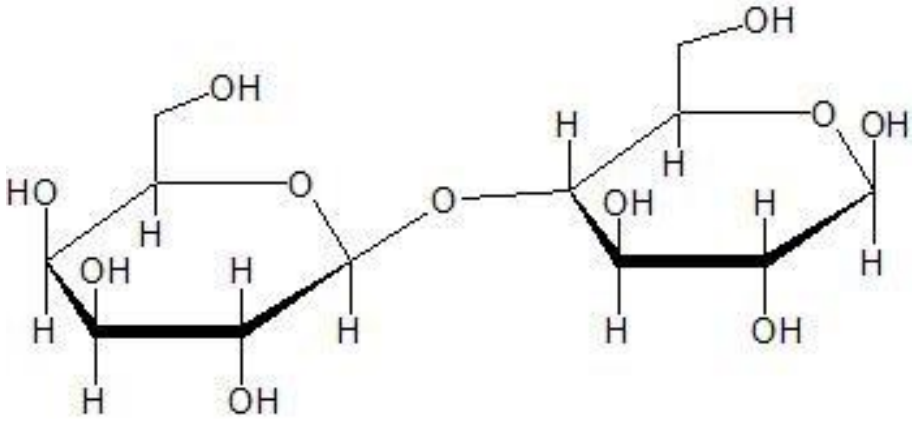


D)



i. Is it a furanose or a pyranose?
ii. Is it an alpha or beta anomer?
iii. Circle the anomeric carbon.

(04) Some people cannot digest the disaccharide lactose. The term for this is known as lactose intolerance. Lactose is shown below. Answer the following questions.

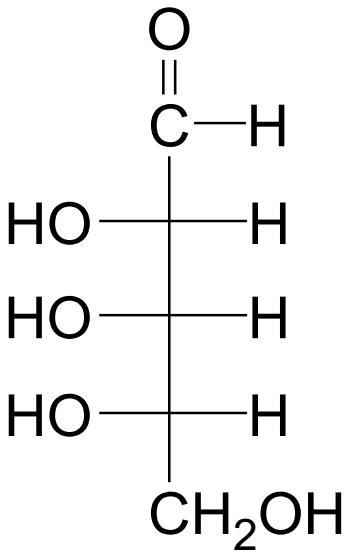


I. What is the name of the enzyme that would be required for someone to be able to digest lactose?
II. Classify lactose as a mono-, di-, oligo-, or polysaccharide.
III. Label anomeric carbons by circling them. Is lactose a reducing sugar?
IV. Draw an arrow pointing to the glycosidic bond. Is the glycosidic bond connected to both anomeric carbons?
V. Classify the glycosidic bonds using the alpha or beta-(#,#) format.
VI. If the glycosidic bond is hydrolyzed, what are the names of the monosaccharides produced. Remember to include the alpha or beta classification for the anomeric carbon.

QUIZ

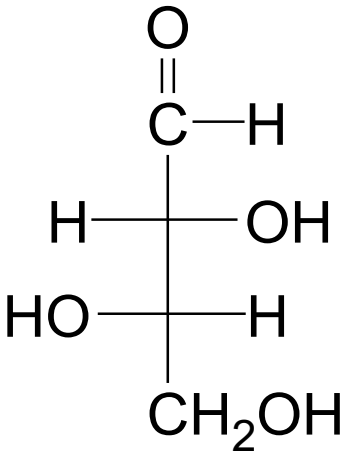
Identify each as the D or L isomer.

A.



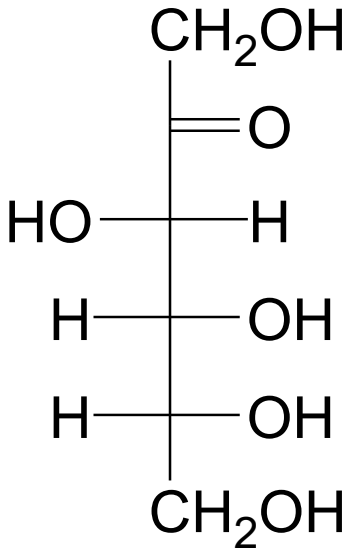
__-ribose

B.



__- threose

C.



__- fructose

QUIZ

Identify the monosaccharides in each of the following disaccharides:

A. lactose

(1) *α*-D-glucose (2) *β*-D-fructose (3) *β*-D-galactose

B. maltose

(1) *α*-D-glucose (2) *β*-D-fructose (3) *β*-D-galactose

C. sucrose

(1) *α*-D-glucose (2) *β*-D-fructose (3) *β*-D-galactose

SAMPLE QUESTIONS

1. Which of the following monosaccharides is the majority found in the human body?

(a) D-type
(b) L-type
(c) Both L and D-types
(d) None of the above

2. Which of the following is the most abundant biomolecule on the earth?

(a) Lipids
(b) Proteins
(c) Carbohydrates
(d) Nucleic acids.

3. Which of the following is an example of Epimers?

(a) Glucose and Ribose
(b) Glucose and Galactose
(c) Galactose, Mannose and Glucose
(d) Glucose, Ribose and Mannose

4. Class of carbohydrate which cannot be hydrolysed further, is known as?

a) Disaccharides
b) Polysaccharides
c) Proteoglycan
d) Monosaccharide

5. Which class of carbohydrates is considered as non-sugar?

a) Monosaccharides
b) Disaccharides
c) Polysaccharides
d) Oligosaccharides

6. When aldoses oxidize under proper conditions, they may form.

a) Aldonic acid
b) Saccharic acids
c) Uronic acid
d) All of these

7. Glucose and mannose are epimers. This means that

A. they are mirror images of each other
B. one is a pyranose, the other a furanose
C. they rotate plane-polarized light in opposite directions
D. they differ only in the configuration of one carbon atom.

8. In order that a compound possesses optical activity it must be

A. Colored
B. Symmetrical
C. Inorganic
D. Asymmetric

9. Which of the following is NOT true about monosaccharides?

A. Are composed of two units joined by a glycosidic bond.
B. Monosaccharides are the simplest form of carbohydrates
C. Can not be broken down into a simpler sugar
D. Serve as the building blocks for more complex carbohydrates

10. The minimum number of carbons in monosaccharide is

A.1
B.2
C.3
D.4

11. Majority of the monosaccharides found in the human body are of

A. L-type
B. D-type
C. DL-types
D. None of the above

12. The α and β forms of glucose are called

a) Isomers
b) Epimers
c) Anomers
d) Enantiomers

Fill in the Blanks

1.Monosaccharides are the _____ units of carbohydrates. Glucose is an example of an _____ sugar.

2.Fructose is a _____ sugar.

3. The cyclic form of glucose can be either α or β depending on the position of the _____ group.

4.Lactose is composed of glucose and _____.

5.The bond between two monosaccharides is called a _____ bond.

6.Sucrose is a _____ sugar.

7.Maltose is formed from two units of _____ glucose.

True or False

1.Glucose and fructose are both hexoses.
2.Lactose is a non-reducing sugar.
3.Epimers differ in configuration around only one specific carbon atom.
4.In the Fischer projection, the D and L forms of sugars are based on the orientation of the hydroxyl group on the penultimate carbon.
5.Sucrose gives a positive Benedict's test.
6.Maltose can be hydrolyzed into glucose and galactose.
7. Monosaccharides can be classified based on the number of carbon atoms and the type of carbonyl group.
8.In α-D-glucose, the –OH group on carbon 1 is below the plane of the ring.
9.Disaccharides are formed by dehydration synthesis.